

CHAPTER 5

Applications of Contour Integration

Chapter Outline

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5.1 INTRODUCTION

The Cauchy's residue theorem is very important in the development and applications of the theory of functions of a complex variable. It is a powerful tool to evaluate line integrals of analytic functions over closed curves. The residue theorem can be applied to real definite integrals. These real integrals are evaluated by expressing them in terms of complex functions over a suitable contour. The process of evaluation of these integrals is called *contour integration*.

5.2 EVALUATION OF A REAL DEFINITE INTEGRAL OF A RATIONAL FUNCTION OF $\cos \theta$ AND $\sin \theta$

Let
$$I = \int_0^{2\pi} f(\sin \theta, \cos \theta) d\theta$$

where $f(\sin \theta, \cos \theta)$ is a rational function of $\sin \theta$ and $\cos \theta$.

To evaluate the integral, putting $z = e^{i\theta}$,

$$dz = i e^{i\theta} d\theta = i z d\theta$$

$$\cos \theta = \frac{1}{2} \left(z + \frac{1}{z} \right) = \frac{z^2 + 1}{2z}$$

$$\sin \theta = \frac{1}{2i} \left(z - \frac{1}{z} \right) = \frac{z^2 - 1}{2iz}$$

$$|z| = |e^{i\theta}| = 1$$

Substituting these values in the integral,

$$I = \oint_C f(z) dz$$

where C is a unit circle $|z| = 1$

By Cauchy's residue theorem,

$$I = 2\pi i \text{ (sum of residues at all poles lying inside the circle } |z| = 1 \text{)}$$

Example 1

Evaluate $\int_0^{2\pi} \frac{d\theta}{5 - \sin \theta}$ using contour integration.

Solution

(i) Let $I = \int_0^{2\pi} \frac{d\theta}{5 - \sin \theta}$

Consider the contour C as the unit circle $|z| = 1$ (Fig. 5.1).

$$\text{Let } z = e^{i\theta}, d\theta = \frac{dz}{iz}, \sin \theta = \frac{z^2 - 1}{2iz}$$

$$\begin{aligned} \text{(ii)} \quad I &= \int_C \frac{\frac{dz}{iz}}{5 - \left(\frac{z^2 - 1}{2iz} \right)} \\ &= \int_C \frac{2}{-z^2 + 10iz + 1} dz \\ &= -2 \int_C \frac{1}{z^2 - 10iz - 1} dz \end{aligned}$$

(iii) Let $f(z) = \frac{1}{z^2 - 10iz - 1}$

The poles are given by

$$z^2 - 10iz - 1 = 0$$

$$z = \frac{10i \pm \sqrt{(-10i)^2 + 4}}{2}$$

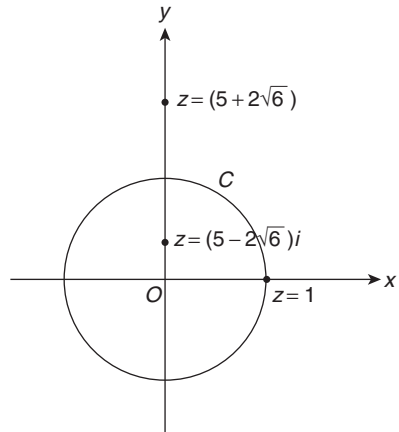


Fig. 5.1

...(1)

$$\begin{aligned}
 &= \frac{10i \pm \sqrt{(-96)}}{2} \\
 &= 5i \pm 2\sqrt{6}i \\
 &= (5 \pm 2\sqrt{6})i
 \end{aligned}$$

$$\therefore f(z) = \frac{1}{\left[z - (5 + 2\sqrt{6})i \right] \left[z - (5 - 2\sqrt{6})i \right]}$$

(iv) For $z = (5 + 2\sqrt{6})i$, $|z| = |(5 + 2\sqrt{6})i| = 9.89 > 1$

Hence, $z = (5 + 2\sqrt{6})i$ lies outside C .

For $z = (5 - 2\sqrt{6})i$, $|z| = |(5 - 2\sqrt{6})i| = 0.10 < 1$

Hence, $z = (5 - 2\sqrt{6})i$ lies inside C .

$$\begin{aligned}
 \text{(v) Res } [f(z); z = (5 - 2\sqrt{6})i] &= \lim_{z \rightarrow (5 - 2\sqrt{6})i} [z - (5 - 2\sqrt{6})i] f(z) \\
 &= \lim_{z \rightarrow (5 - 2\sqrt{6})i} \frac{1}{z - (5 + 2\sqrt{6})i} \\
 &= \frac{1}{(5 - 2\sqrt{6})i - (5 + 2\sqrt{6})i} \\
 &= -\frac{1}{4\sqrt{6}i}
 \end{aligned}$$

(vi) By Cauchy's residue theorem,

$$\begin{aligned}
 \int_C f(z) dz &= 2\pi i \text{ (sum of residues)} \\
 \int_C \frac{dz}{z^2 - 10iz - 1} &= 2\pi i \left(-\frac{1}{4\sqrt{6}i} \right) \\
 &= -\frac{\pi}{2\sqrt{6}}
 \end{aligned}$$

Substituting in Eq. (1),

$$\begin{aligned}
 I &= -2 \left(-\frac{\pi}{2\sqrt{6}} \right) \\
 &= \frac{\pi}{\sqrt{6}}
 \end{aligned}$$

Example 2

Evaluate $\int_0^{2\pi} \frac{d\theta}{13+5\sin\theta}$ using contour integration.

Solution

(i) Let $I = \int_0^{2\pi} \frac{d\theta}{13+5\sin\theta}$

Consider the contour C as the unit circle $|z| = 1$ (Fig. 5.2).

$$z = e^{i\theta}, d\theta = \frac{dz}{iz}, \sin\theta = \frac{z^2 - 1}{2iz}$$

$$\begin{aligned} \text{(ii) } I &= \int_C \frac{\frac{dz}{iz}}{13+5\left(\frac{z^2-1}{2iz}\right)} \\ &= 2 \int_C \frac{dz}{5z^2 + 26iz - 5} \end{aligned} \quad \dots(1)$$

(iii) Let $f(z) = \frac{1}{5z^2 + 26iz - 5}$

The poles are given by

$$\begin{aligned} z &= \frac{-26i \pm \sqrt{(26i)^2 - 4(5)(-5)}}{10} \\ &= \frac{-26i \pm \sqrt{-576}}{10} \\ &= \frac{-26i \pm 24i}{10} \\ &= -\frac{i}{5}, -5i \end{aligned}$$

$$\therefore f(z) = \frac{1}{5\left(z + \frac{i}{5}\right)(z + 5i)}$$

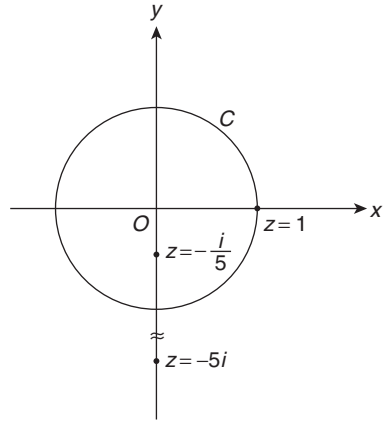


Fig. 5.2

$$\left[\because ax^2 + bx + c = a(x - \alpha)(x - \beta) \right]$$

(iv) For $z = -\frac{i}{5}$, $|z| = \left| -\frac{i}{5} \right| = \frac{1}{5} < 1$

Hence, $z = -\frac{i}{5}$ lies inside C .

For $z = -5i$, $|z| = |-5i| = 5 > 1$

Hence, $z = -5i$ lies outside C .

$$\begin{aligned}
 \text{(v) } \operatorname{Res} \left[f(z); z = -\frac{i}{5} \right] &= \lim_{z \rightarrow -\frac{i}{5}} \left[\left(z + \frac{i}{5} \right) f(z) \right] \\
 &= \lim_{z \rightarrow -\frac{i}{5}} \frac{1}{5(z+5i)} \\
 &= \frac{1}{5 \left(-\frac{i}{5} + 5i \right)} \\
 &= \frac{1}{24i}
 \end{aligned}$$

(vi) By Cauchy's residue theorem,

$$\begin{aligned}
 \int_C f(z) \, dz &= 2\pi i \text{ (sum of residues)} \\
 \int_C \frac{dz}{5z^2 + 26iz - 5} &= 2\pi i \left(\frac{1}{24i} \right) \\
 &= \frac{\pi}{12}
 \end{aligned}$$

Substituting in Eq. (1),

$$\begin{aligned}
 I &= 2 \left(\frac{\pi}{12} \right) \\
 &= \frac{\pi}{6}
 \end{aligned}$$

Example 3

Evaluate $\int_0^\pi \frac{1}{17 - 8 \cos \theta} d\theta$, by integrating around a unit circle.

Solution

$$\text{(i) Let } I = \int_0^{2\pi} \frac{1}{17 - 8 \cos \theta} d\theta$$

Consider the contour C as the unit circle $|z| = 1$ (Fig. 5.3).

$$\text{Let } z = e^{i\theta}, \quad d\theta = \frac{dz}{iz}, \quad \cos \theta = \frac{z^2 + 1}{2z}$$

$$\begin{aligned} \text{(ii)} \quad I &= \int_C \frac{1}{17 - 8\left(\frac{z^2 + 1}{2z}\right)} \frac{dz}{iz} \\ &= \frac{1}{i} \int_C \frac{1}{17z - 4z^2 - 4} dz \\ &= -\frac{1}{i} \int_C \frac{1}{4z^2 - 17z + 4} dz \end{aligned}$$

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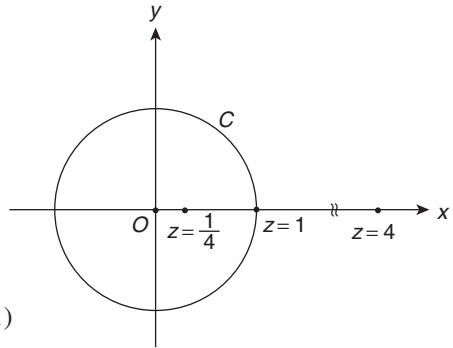


Fig. 5.3

$$\text{(iii)} \quad \text{Let } f(z) = \frac{1}{4z^2 - 17z + 4}$$

The poles are given by

$$\begin{aligned} 4z^2 - 17z + 4 &= 0 \\ z &= \frac{17 \pm \sqrt{289 - 64}}{8} \\ &= \frac{17 \pm \sqrt{225}}{8} \\ &= \frac{17 \pm 15}{8} \\ &= 4, \frac{1}{4} \end{aligned}$$

$$\therefore f(z) = \frac{1}{4(z-4)\left(z-\frac{1}{4}\right)} \quad \left[\because ax^2 + bx + c = a(x-\alpha)(x-\beta) \right]$$

$$\text{(iv)} \quad \text{For } z = 4, |z| = |4| = 4 > 1$$

Hence, $z = 4$ lies outside C .

$$\text{For } z = \frac{1}{4}, |z| = \left|\frac{1}{4}\right| = \frac{1}{4} < 1$$

Hence, $z = \frac{1}{4}$ lies inside C .

$$\begin{aligned} \text{(v)} \quad \text{Res} \left[f(z); z = \frac{1}{4} \right] &= \lim_{z \rightarrow \frac{1}{4}} \left[\left(z - \frac{1}{4} \right) f(z) \right] \\ &= \lim_{z \rightarrow \frac{1}{4}} \frac{1}{4(z-4)} \end{aligned}$$

$$= \frac{1}{4\left(\frac{1}{4} - 4\right)}$$

$$= -\frac{1}{15}$$

(vi) By Cauchy's residue theorem,

$$\int_C f(z) dz = 2\pi i \text{ (sum of residues)}$$

$$\int_C \frac{1}{4z^2 - 17z + 4} dz = 2\pi i \left(-\frac{1}{15}\right)$$

$$= -\frac{2\pi i}{15}$$

Substituting in Eq. (1),

$$I = -\frac{1}{i} \left(-\frac{2\pi i}{15}\right)$$

$$= \frac{2\pi}{15}$$

$$\int_0^{2\pi} \frac{1}{17 - 8 \cos \theta} d\theta = \frac{2\pi}{15}$$

$$2 \int_0^{\pi} \frac{1}{17 - 8 \cos \theta} d\theta = \frac{2\pi}{15} \quad \left[\because \int_0^{2a} f(x) dx = 2 \int_0^a f(x) dx \text{ if } f(2a - x) = f(x) \right]$$

$$\int_0^{\pi} \frac{1}{17 - 8 \cos \theta} d\theta = \frac{\pi}{15}$$

Example 4

Evaluate $\int_0^{2\pi} \frac{d\theta}{2 + \cos \theta}$ using contour integration.

Solution

(i) Let $I = \int_0^{2\pi} \frac{d\theta}{2 + \cos \theta}$

Consider the contour C as the unit circle $|z| = 1$ (Fig. 5.4).

Let $z = e^{i\theta}$, $d\theta = \frac{dz}{iz}$, $\cos \theta = \frac{z^2 + 1}{2z}$

$$\begin{aligned}
 \text{(ii) } I &= \int_C \frac{\frac{dz}{iz}}{2 + \frac{z^2+1}{2z}} \\
 &= \frac{2}{i} \int_C \frac{dz}{4z + z^2 + 1} \\
 &= \frac{2}{i} \int_C \frac{dz}{z^2 + 4z + 1} \quad \dots(1)
 \end{aligned}$$

$$\text{(iii) Let } f(z) = \frac{1}{z^2 + 4z + 1}$$

The poles are given by

$$\begin{aligned}
 z^2 - 4z + 1 &= 0 \\
 z &= \frac{-4 \pm \sqrt{16-4}}{2} \\
 &= \frac{-4 \pm 2\sqrt{3}}{2} \\
 &= -2 \pm \sqrt{3}
 \end{aligned}$$

$$\therefore f(z) = \frac{1}{[z - (-2 + \sqrt{3})][z - (-2 - \sqrt{3})]}$$

$$\text{(iv) For } z = -2 + \sqrt{3}, |z| = |-2 + \sqrt{3}| = 0.27 < 1$$

Hence, $z = -2 + \sqrt{3}$ lies inside C .

$$\text{For } z = -2 - \sqrt{3}, |z| = |-2 - \sqrt{3}| = 3.73 > 1$$

Hence, $z = -2 - \sqrt{3}$ lies outside C .

$$\begin{aligned}
 \text{(v) Res } [f(z); z = -2 + \sqrt{3}] &= \lim_{z \rightarrow -2 + \sqrt{3}} [(z + 2 - \sqrt{3})f(z)] \\
 &= \lim_{z \rightarrow -2 + \sqrt{3}} \frac{1}{z + 2 + \sqrt{3}} \\
 &= \frac{1}{-2 + \sqrt{3} + 2 + \sqrt{3}} \\
 &= \frac{1}{2\sqrt{3}}
 \end{aligned}$$

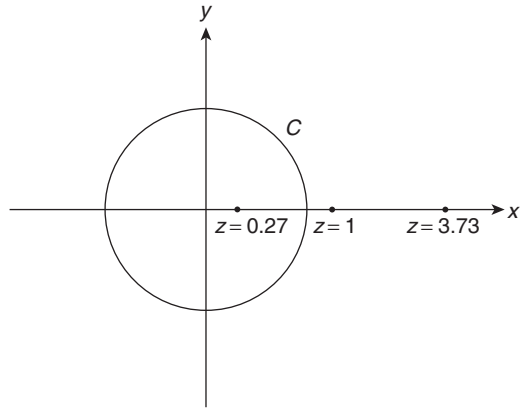


Fig. 5.4

(vi) By Cauchy's residue theorem,

$$\begin{aligned} \int_C f(z) dz &= 2\pi i \text{ (sum of residues)} \\ \int_C \frac{dz}{z^2 + 4z + 1} &= 2\pi i \left(\frac{1}{2\sqrt{3}} \right) \\ &= \frac{\pi i}{\sqrt{3}} \end{aligned}$$

Substituting in Eq. (1),

$$\begin{aligned} I &= \frac{2}{i} \left(\frac{\pi i}{\sqrt{3}} \right) \\ &= \frac{2\pi}{\sqrt{3}} \end{aligned}$$

Example 5

Evaluate a real integral $\int_0^{2\pi} \frac{1}{(2 + \cos \theta)^2} d\theta$ using residue. [Summer 2013]

Solution

(i) Let $I = \int_0^{2\pi} \frac{1}{(2 + \cos \theta)^2} d\theta$

Consider the contour C as the unit circle $|z| = 1$ (Fig. 5.5).

Let $z = e^{i\theta}$, $d\theta = \frac{dz}{iz}$, $\cos \theta = \frac{z^2 + 1}{2z}$

$$\begin{aligned} \text{(ii) } I &= \int_C \frac{1}{\left(2 + \frac{z^2 + 1}{2z}\right)^2} \frac{dz}{iz} \\ &= \frac{4}{i} \int_C \frac{z}{(4z + z^2 + 1)^2} dz \\ &= \frac{4}{i} \int_C \frac{z}{(z^2 + 4z + 1)^2} dz \quad \dots(1) \end{aligned}$$

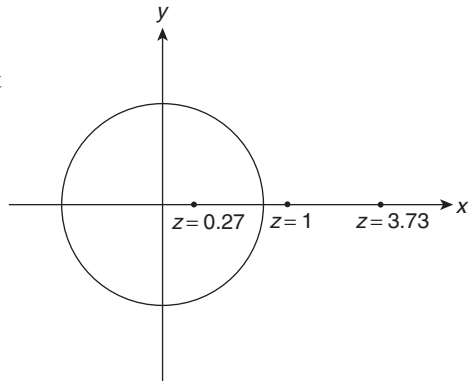


Fig. 5.5

(iii) Let $f(z) = \frac{z}{(z^2 + 4z + 1)^2}$

The poles are given by

$$\begin{aligned}(z^2 + 4z + 1)^2 &= 0 \\ z &= \frac{-4 \pm \sqrt{16 - 4}}{2} \\ &= -2 \pm \sqrt{3}\end{aligned}$$

$$\begin{aligned}\therefore f(z) &= \frac{z}{[z - (-2 + \sqrt{3})]^2 [z - (-2 - \sqrt{3})]^2} \\ &= \frac{z}{(z + 2 - \sqrt{3})^2 (z + 2 + \sqrt{3})^2}\end{aligned}$$

(iv) For $z = -2 + \sqrt{3}$, $|z| = |-2 + \sqrt{3}| = 0.27 < 1$

Hence, $z = -2 + \sqrt{3}$ lies inside C .

For $z = -2 - \sqrt{3}$, $|z| = |-2 - \sqrt{3}| = 3.73 > 1$

Hence, $z = -2 - \sqrt{3}$ lies outside C .

(v) $z = -2 + \sqrt{3}$ is a pole of order 2.

$$\begin{aligned}\text{Res}[f(z); z = -2 + \sqrt{3}] &= \frac{1}{(2-1)!} \lim_{z \rightarrow (-2 + \sqrt{3})} \frac{d}{dz} \left[(z + 2 - \sqrt{3})^2 f(z) \right] \\ &= \lim_{z \rightarrow (-2 + \sqrt{3})} \frac{d}{dz} \frac{z}{(z + 2 + \sqrt{3})^2} \\ &= \lim_{z \rightarrow (-2 + \sqrt{3})} \left[-\frac{2z}{(z + 2 + \sqrt{3})^3} + \frac{1}{(z + 2 + \sqrt{3})^2} \right] \\ &= -\frac{2(-2 + \sqrt{3})}{(-2 + \sqrt{3} + 2 + \sqrt{3})^3} + \frac{1}{(-2 + \sqrt{3} + 2 + \sqrt{3})^2} \\ &= -\frac{(-4 + 2\sqrt{3})}{(2\sqrt{3})^3} + \frac{1}{(2\sqrt{3})^2} \\ &= -\frac{(-2 + \sqrt{3})}{12\sqrt{3}} + \frac{1}{12} \\ &= \frac{2 - \sqrt{3} + \sqrt{3}}{12\sqrt{3}} \\ &= \frac{2}{12\sqrt{3}} \\ &= \frac{1}{6\sqrt{3}}\end{aligned}$$

(vi) By Cauchy's residue theorem,

$$\begin{aligned} \int_C f(z) dz &= 2\pi i (\text{sum of residues}) \\ \int_C \frac{z}{(z^2 + 4z + 1)^2} dz &= 2\pi i \left(\frac{1}{6\sqrt{3}} \right) \\ &= \frac{\pi i}{3\sqrt{3}} \end{aligned}$$

Substituting in Eq. (1),

$$\begin{aligned} I &= \frac{4}{i} \left(\frac{\pi i}{3\sqrt{3}} \right) \\ &= \frac{4\pi}{3\sqrt{3}} \end{aligned}$$

Example 6

Evaluate $\int_0^{2\pi} \frac{\cos 3\theta}{5 - 4 \cos \theta} d\theta$ using contour integration.

Solution

(i) Let $I = \int_0^{2\pi} \frac{\cos 3\theta}{5 - 4 \cos \theta} d\theta$

Consider the contour C as the unit circle $|z| = 1$ (Fig. 5.6).

$$z = e^{i\theta}, d\theta = \frac{dz}{iz}, \cos \theta = \frac{z^2 + 1}{2z}$$

$$z^3 = e^{3i\theta} = \cos 3\theta + i \sin 3\theta$$

$$\cos 3\theta = \text{RP } z^3$$

(ii) $I = \text{RP} \int_C \frac{z^3}{5 - 4 \left(\frac{z^2 + 1}{2z} \right)} \frac{dz}{iz}$

$$= \text{RP} \frac{1}{i} \int_C \frac{z^3}{5z - 2z^2 - 2} dz$$

$$= \text{RP} \left(-\frac{1}{i} \right) \int_C \frac{z^3}{2z^2 - 5z + 2} dz \quad \dots(1)$$

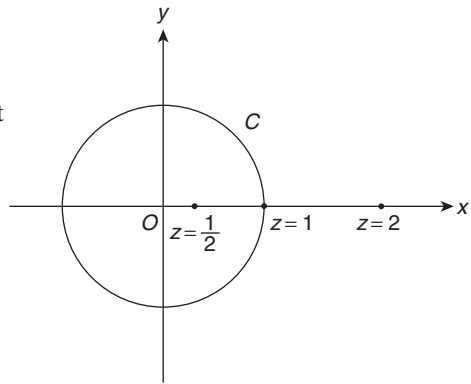


Fig. 5.6

(iii) Let $f(z) = \frac{z^3}{2z^2 - 5z + 2}$

The poles are given by

$$2z^2 - 5z + 2 = 0$$

$$2z^2 - 4z - z + 2 = 0$$

$$2z(z-2) - 1(z-2) = 0$$

$$(z-2)(2z-1) = 0$$

$$z = 2, \quad \frac{1}{2}$$

$$\therefore f(z) = \frac{z^3}{2(z-2)\left(z - \frac{1}{2}\right)} \quad [\because ax^2 + bx + c = a(x - \alpha)(x - \beta)]$$

(iv) For $z = 2, |z| = |2| = 2 > 1$

Hence, $z = 2$ lies outside C .

For $z = \frac{1}{2}, |z| = \left|\frac{1}{2}\right| = \frac{1}{2} < 1$

Hence, $z = \frac{1}{2}$ lies inside C .

(v) $\text{Res} \left[f(z); z = \frac{1}{2} \right] = \lim_{z \rightarrow \frac{1}{2}} \left[\left(z - \frac{1}{2} \right) f(z) \right]$

$$= \lim_{z \rightarrow \frac{1}{2}} \frac{z^3}{2(z-2)}$$

$$= \frac{\frac{1}{8}}{2\left(\frac{1}{2} - 2\right)}$$

$$= -\frac{1}{24}$$

(vi) By Cauchy's residue theorem,

$$\int_C f(z) dz = 2\pi i (\text{sum of residues})$$

$$\begin{aligned} \int_C \frac{z^3 dz}{2z^2 - 5z + 2} &= 2\pi i \left(-\frac{1}{24} \right) \\ &= -\frac{\pi i}{12} \end{aligned}$$

Substituting in Eq. (1),

$$\begin{aligned} I &= \text{RP} \left(-\frac{1}{i} \right) \left(\frac{-\pi i}{12} \right) \\ &= \frac{\pi}{12} \end{aligned}$$

Example 7

Evaluate $\int_0^{2\pi} \frac{1+2\cos\theta}{5+4\cos\theta} d\theta$ using contour integration.

Solution

(i) Let $I = \int_0^{2\pi} \frac{1+2\cos\theta}{5+4\cos\theta} d\theta$

Consider the contour C as the unit circle $|z| = 1$ (Fig. 5.7).

Let $z = e^{i\theta}$, $d\theta = \frac{dz}{iz}$, $\cos\theta = \frac{z^2+1}{2z}$,

$$\begin{aligned} \text{(ii) } I &= \int_C \frac{1+2\left(\frac{z^2+1}{2z}\right)}{5+4\left(\frac{z^2+1}{2z}\right)} \frac{dz}{iz} \\ &= \frac{1}{i} \int_C \frac{z+z^2+1}{z(5z+2z^2+2)} dz \\ &= \frac{1}{i} \int_C \frac{z^2+z+1}{z(2z^2+5z+2)} dz \quad \dots(1) \end{aligned}$$

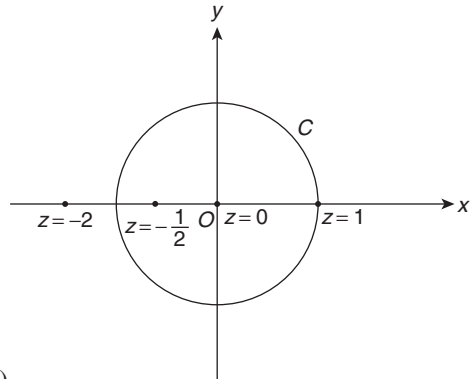


Fig. 5.7

(iii) Let $f(z) = \frac{z^2+z+1}{z(2z^2+5z+2)}$

The poles are given by

$$z(2z^2+5z+2) = 0$$

$$z(z+2)(2z+1) = 0$$

$$z = 0, -2, -\frac{1}{2}$$

$$\therefore f(z) = \frac{z^2+z+1}{2z(z+2)\left(z+\frac{1}{2}\right)}$$

$$[\because ax^2+bx+c = a(x-\alpha)(x-\beta)]$$

(iv) For $z = 0, |z| = 0 < 1$

Hence, $z = 1$ lies inside C .

For, $z = -2, |z| = |-2| = 2 > 1$

Hence, $z = -2$ lies outside C .

For, $z = -\frac{1}{2}, |z| = \left|-\frac{1}{2}\right| = \frac{1}{2} < 1$

Hence, $z = -\frac{1}{2}$ lies inside C .

$$\begin{aligned}
 \text{(v) Res } [f(z); z = 0] &= \lim_{z \rightarrow 0} zf(z) \\
 &= \lim_{z \rightarrow 0} \frac{z^2 + z + 1}{2(z+2) \left(z + \frac{1}{2}\right)} \\
 &= \frac{0+0+1}{2(0+2) \left(0 + \frac{1}{2}\right)} \\
 &= \frac{1}{2} \\
 \text{Res } \left[f(z); z = -\frac{1}{2} \right] &= \lim_{z \rightarrow -\frac{1}{2}} \left(z + \frac{1}{2} \right) f(z) \\
 &= \lim_{z \rightarrow -\frac{1}{2}} \frac{z^2 + z + 1}{2z(z+2)} \\
 &= \frac{\frac{1}{4} - \frac{1}{2} + 1}{2 \left(-\frac{1}{2} \right) \left(-\frac{1}{2} + 2 \right)} \\
 &= -\frac{1}{2}
 \end{aligned}$$

(vi) By Cauchy's residue theorem,

$$\begin{aligned}
 \int_C f(z) dz &= 2\pi i (\text{sum of residues}) \\
 \int_C \frac{z^2 + z + 1}{z(2z^2 + 5z + 2)} dz &= 2\pi i \left(\frac{1}{2} - \frac{1}{2} \right) \\
 &= 0
 \end{aligned}$$

Substituting in Eq. (1),

$$I = \frac{1}{i}(0) = 0$$

Example 8

Evaluate $\int_0^{2\pi} \frac{d\theta}{3 - 2\cos \theta + \sin \theta}$ using contour integration.

Solution

(i) Let $I = \int_0^{2\pi} \frac{d\theta}{3 - 2\cos \theta + \sin \theta}$

Consider the contour C as the unit circle $|z| = 1$ (Fig. 5.8).

$$z = e^{i\theta}, d\theta = \frac{dz}{iz}, \cos \theta = \frac{z^2 + 1}{2z}, \sin \theta = \frac{z^2 - 1}{2iz},$$

$$\begin{aligned} \text{(ii) } I &= \int_C \frac{\frac{dz}{iz}}{3 - 2\left(\frac{z^2 + 1}{2z}\right) + \left(\frac{z^2 - 1}{2iz}\right)} \\ &= \int_C \frac{\frac{dz}{iz}}{3 - \frac{2z^2 + 2}{2z} + \frac{z^2 - 1}{2iz}} \\ &= \int_C \frac{2dz}{6iz - (2z^2 + 2)i + (z^2 - 1)} \\ &= 2 \int \frac{dz}{(1 - 2i)z^2 + 6iz - (2i + 1)} \end{aligned}$$

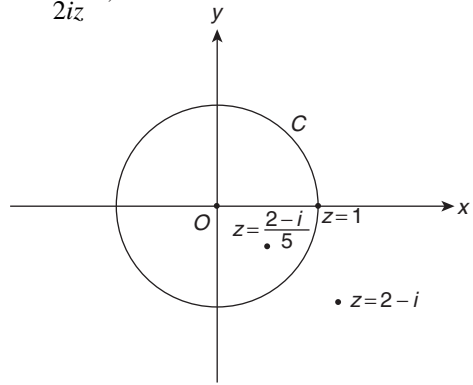


Fig. 5.8

(iii) Let $f(z) = \frac{1}{(1 - 2i)z^2 + 6iz - (2i + 1)}$

The poles are given by

$$(1 - 2i)z^2 + 6iz - (2i + 1) = 0$$

$$\begin{aligned} z &= \frac{-6i \pm \sqrt{(6i)^2 + 4(1 - 2i)(2i + 1)}}{2(1 - 2i)} \\ &= \frac{-6i \pm \sqrt{-36 + 4(2i + 1 + 4 - 2i)}}{2(1 - 2i)} \\ &= \frac{-6i \pm \sqrt{-36 + 20}}{2(1 - 2i)} \\ &= \frac{-6i \pm 4i}{2(1 - 2i)} \end{aligned}$$

...(1)

$$= \frac{i}{1-2i}, \frac{-5i}{1-2i}$$

$$= \frac{2-i}{5}, 2-i$$

$$\therefore f(z) = \frac{1}{(1-2i)\left[z-(2-i)\right]\left[z-\left(\frac{2-i}{5}\right)\right]} \quad [\because ax^2+bx+c=a(x-\alpha)(x-\beta)]$$

(iv) For $z = 2-i$, $|z| = |2-i| = \sqrt{5} > 1$

Hence, $z = 2-i$ lies outside C .

$$\text{For } z = \frac{2-i}{5}, |z| = \left|\frac{2-i}{5}\right| = \frac{\sqrt{5}}{5} = 0.45 < 1$$

Hence, $z = \frac{2-i}{5}$ lies inside C .

$$\begin{aligned} \text{(v) Res}\left[f(z); z = \frac{2-i}{5}\right] &= \lim_{z \rightarrow \frac{2-i}{5}} \left[\left(z - \frac{2-i}{5}\right) f(z) \right] \\ &= \lim_{z \rightarrow \frac{2-i}{5}} \frac{1}{(1-2i)(z-2+i)} \\ &= \frac{1}{(1-2i)\left(\frac{2-i}{5} - 2 + i\right)} \\ &= \frac{1+2i}{5\left(\frac{-8+4i}{5}\right)} \\ &= \frac{1+2i}{4i(2i+1)} \\ &= \frac{1}{4i} \end{aligned}$$

(vi) By Cauchy's residue theorem,

$$\int_C f(z) dz = 2\pi i \text{ (sum of residues)}$$

$$\int_C \frac{dz}{(1-2i)z^2 + 6iz - (2i+1)} = 2\pi i \left(\frac{1}{4i} \right)$$

$$= \frac{\pi}{2}$$

Substituting in Eq. (1),

$$I = 2 \left(\frac{\pi}{2} \right) = \pi$$

Example 9

Evaluate $\int_0^{2\pi} \frac{d\theta}{1 - 2a \cos \theta + a^2}$, ($0 < a < 1$) using contour integration.

Solution

(i) Let $I = \int_0^{2\pi} \frac{d\theta}{1 - 2a \cos \theta + a^2}$

Consider the contour C as the unit circle $|z| = 1$ (Fig. 5.9).

Let $z = e^{i\theta}$, $d\theta = \frac{dz}{iz}$, $\cos \theta = \frac{z^2 + 1}{2z}$,

$$\begin{aligned} \text{(ii) } I &= \int_C \frac{\frac{dz}{iz}}{1 - 2a \left(\frac{z^2 + 1}{2z} \right) + a^2} \\ &= \frac{1}{i} \int_C \frac{dz}{z - az^2 - a + a^2 z} \\ &= \frac{1}{i} \int_C \frac{dz}{-az^2 + (a^2 + 1)z - a} \\ &= -\frac{1}{i} \int_C \frac{dz}{az^2 - (a^2 + 1)z + a} \quad \dots(1) \end{aligned}$$

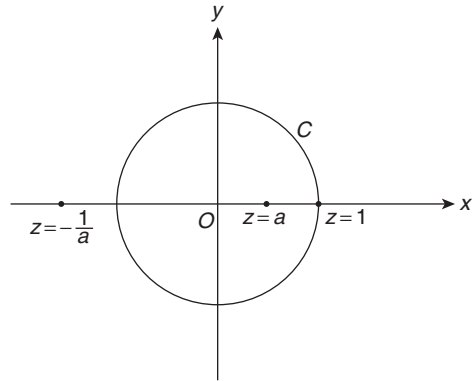


Fig. 5.9

(iii) Let $f(z) = \frac{1}{az^2 - (a^2 + 1)z + a}$

The poles are given by

$$z^2 - \left(\frac{a^2 + 1}{a} \right) z + 1 = 0$$

$$\begin{aligned} z &= \frac{a^2 + 1 \pm \sqrt{(a^2 + 1)^2 - 4a^2}}{2a} \\ &= \frac{a^2 + 1 \pm \sqrt{a^4 + 2a^2 + 1 - 4a^2}}{2a} \end{aligned}$$

$$\begin{aligned}
&= \frac{a^2 + 1 \pm \sqrt{(a^2 - 1)^2}}{2a} \\
&= \frac{a^2 + 1 \pm (a^2 - 1)}{2a} \\
&= a, \frac{1}{a}
\end{aligned}$$

$$\therefore f(z) = \frac{1}{a(z-a)\left(z - \frac{1}{a}\right)} \quad [\because ax^2 + bx + c = a(x - \alpha)(x - \beta)]$$

(iv) For $z = a, |z| = |a| = a < 1$

Hence, $z = a$ lies inside C .

For $z = \frac{1}{a}, |z| = \left|\frac{1}{a}\right| = \frac{1}{a} > 1$

Hence, $z = \frac{1}{a}$ lies outside C .

$$\begin{aligned}
\text{(v) Res}[f(z); z = a] &= \lim_{z \rightarrow a} [(z-a)f(z)] \\
&= \lim_{z \rightarrow a} \frac{1}{a\left(z - \frac{1}{a}\right)} \\
&= \frac{1}{a\left(a - \frac{1}{a}\right)} \\
&= \frac{1}{a^2 - 1}
\end{aligned}$$

(vi) By Cauchy's residue theorem,

$$\begin{aligned}
\int_C f(z) dz &= 2\pi i (\text{sum of residues}) \\
\int_C \frac{dz}{az^2 - (a^2 + 1)z + a} &= 2\pi i \left(\frac{1}{a^2 - 1} \right)
\end{aligned}$$

Substituting in Eq. (1),

$$\begin{aligned}
I &= -\frac{1}{i} 2\pi i \left(\frac{1}{a^2 - 1} \right) \\
&= -\frac{2\pi}{a^2 - 1} \\
&= \frac{2\pi}{1 - a^2}, 0 < a < 1
\end{aligned}$$

Example 10

Evaluate $\int_0^{2\pi} \frac{\sin^2 \theta}{a + b \cos \theta} d\theta$, $a > b > 0$ using contour integration.

Solution

$$\begin{aligned} \text{(i) Let } I &= \int_0^{2\pi} \frac{\sin^2 \theta}{a + b \cos \theta} d\theta \\ &= \int_0^{2\pi} \frac{1 - \cos 2\theta}{2(a + b \cos \theta)} d\theta \end{aligned}$$

Consider the contour C as the unit circle $|z| = 1$ (Fig. 5.10).

$$z = e^{i\theta}, d\theta = \frac{dz}{iz}, \cos \theta = \frac{z^2 + 1}{2z}$$

$$z^2 = e^{2i\theta} = \cos 2\theta + i \sin 2\theta$$

$$\cos 2\theta = \text{RP } z^2$$

$$1 - \cos 2\theta = \text{RP } (1 - z^2)$$

$$\text{(ii) } I = \text{RP} \int_C \frac{(1 - z^2)^2 \frac{dz}{iz}}{2 \left[a + b \left(\frac{z^2 + 1}{2z} \right) \right]}$$

$$= \text{RP} \frac{1}{i} \int_C \frac{(1 - z^2) dz}{2az + b(z^2 + 1)}$$

$$= \text{RP} \frac{1}{i} \int_C \frac{(1 - z^2) dz}{bz^2 + 2az + b} \quad \dots(1)$$

$$\text{(iii) Let } f(z) = \frac{1 - z^2}{bz^2 - 2az + b}$$

The poles are given by

$$bz^2 + 2az + b = 0$$

$$z = \frac{-2a \pm \sqrt{4a^2 - 4b^2}}{2b}$$

$$= \frac{-2a \pm 2\sqrt{a^2 - b^2}}{2b}$$

$$= \frac{-a \pm \sqrt{a^2 - b^2}}{b}$$

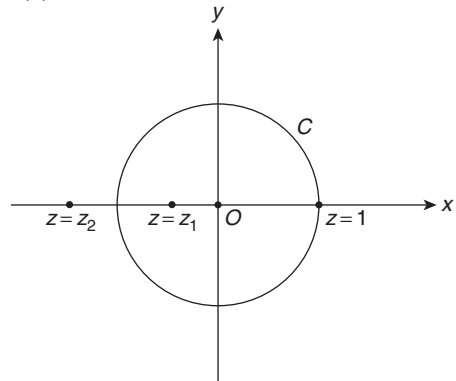


Fig. 5.10

$$\text{Let } z_1 = \frac{-a + \sqrt{a^2 - b^2}}{b}, z_2 = \frac{-a - \sqrt{a^2 - b^2}}{b}$$

$$\therefore f(z) = \frac{1 - z^2}{b(z - z_1)(z - z_2)}$$

$$(iv) \text{ For } z = z_1 = \frac{-a + \sqrt{a^2 - b^2}}{b}$$

$$|z| = |z_1| = \left| \frac{-a + \sqrt{a^2 - b^2}}{b} \right| = \frac{a}{b} - \sqrt{\frac{a^2}{b^2} - 1} < 1 \quad \left[\because a > b, \frac{a}{b} > 1 \right]$$

Hence, $z = z_1$ lies inside C .

$$\text{For } z = z_2 = \frac{-a - \sqrt{a^2 - b^2}}{b}$$

$$|z| = |z_2| = \left| \frac{-a - \sqrt{a^2 - b^2}}{b} \right| = \frac{a}{b} + \sqrt{\frac{a^2}{b^2} - 1} > 1 \quad \left[\because a > b, \frac{a}{b} > 1 \right]$$

Hence, $z = z_2$ lies outside C .

$$(v) \text{ Res}[f(z); z = z_1] = \lim_{z \rightarrow z_1} [(z - z_1)f(z)]$$

$$\begin{aligned} &= \lim_{z \rightarrow z_1} \frac{(1 - z^2)}{b(z - z_2)} \\ &= \frac{1 - z_1^2}{b(z_1 - z_2)} \\ &= \frac{\left[1 - \left(\frac{-a + \sqrt{a^2 - b^2}}{b} \right)^2 \right]}{b \left(\frac{2\sqrt{a^2 - b^2}}{b} \right)} \\ &= \frac{b^2 - \left[a^2 + (a^2 - b^2) - 2a\sqrt{a^2 - b^2} \right]}{2b^2\sqrt{a^2 - b^2}} \\ &= \frac{2(b^2 - a^2) - 2a\sqrt{a^2 - b^2}}{2b^2\sqrt{a^2 - b^2}} \\ &= \frac{-\sqrt{a^2 - b^2} - a}{b^2} \\ &= - \left(\frac{a + \sqrt{a^2 - b^2}}{b^2} \right) \end{aligned}$$

$$(vi) \quad \int_C f(z) dz = 2\pi i \text{ (sum of residues)}$$

$$\int_C \frac{(1-z^2)}{bz^2-2az+b} dz = -2\pi i \left(\frac{a+\sqrt{a^2-b^2}}{b^2} \right)$$

Substituting in Eq. (1),

$$\begin{aligned} I &= \text{RP} \frac{1}{i} \left[-2\pi i \left(\frac{a+\sqrt{a^2-b^2}}{b^2} \right) \right] \\ &= \frac{-2\pi \left(a+\sqrt{a^2-b^2} \right)}{b^2} \end{aligned}$$

EXERCISE 5.1

Evaluate the following integrals using contour integration:

$$1. \quad \int_0^{2\pi} \frac{d\theta}{5+4\cos\theta} \quad \left[\text{Ans.: } \frac{2\pi}{3} \right]$$

$$2. \quad \int_0^{2\pi} \frac{d\theta}{5+4\sin\theta} \quad \left[\text{Ans.: } \frac{2\pi}{3} \right]$$

$$3. \quad \int_0^{2\pi} \frac{\sin\theta}{5+4\cos\theta} d\theta \quad \left[\text{Ans.: } 0 \right]$$

$$4. \quad \int_0^{2\pi} \frac{\cos 2\theta}{5-4\cos\theta} \quad \left[\text{Ans.: } \frac{\pi}{6} \right]$$

$$5. \quad \int_0^{2\pi} \frac{\sin^2\theta}{5-3\cos\theta} d\theta \quad \left[\text{Ans.: } \frac{2\pi}{3} \right]$$

$$6. \quad \int_0^{2\pi} \frac{d\theta}{a+b\sin\theta}, \quad a > b > 0 \quad \left[\text{Ans.: } \frac{2\pi}{\sqrt{a^2-b^2}} \right]$$

$$7. \int_0^{2\pi} \frac{d\theta}{2 + \cos \theta} \quad \left[\text{Ans.: } \frac{2\pi}{\sqrt{3}} \right]$$

$$8. \int_0^{2\pi} \frac{d\theta}{5 - 3 \cos \theta} \quad \left[\text{Ans.: } \frac{\pi}{2} \right]$$

$$9. \int_0^{2\pi} \frac{d\theta}{a + b \cos \theta}, a > b > 0 \quad \left[\text{Ans.: } \frac{2\pi}{\sqrt{a^2 - b^2}} \right]$$

$$10. \int_0^{2\pi} \frac{d\theta}{1 + a \sin \theta}, |a| < 1 \quad \left[\text{Ans.: } \frac{2\pi}{\sqrt{1 - a^2}} \right]$$

$$11. \int_0^{2\pi} \frac{d\theta}{(a + b \cos \theta)^2}, a > b > 0 \quad \left[\text{Ans.: } \frac{2\pi a}{(a^2 - b^2)^{\frac{3}{2}}} \right]$$

$$12. \int_0^{2\pi} \frac{d\theta}{(5 - 4 \cos \theta)^2} \quad \left[\text{Ans.: } \frac{10\pi}{27} \right]$$

5.3 EVALUATION OF IMPROPER REAL INTEGRAL OF A RATIONAL FUNCTION

$$\text{Let } I = \int_{-\infty}^{\infty} f(x) dx$$

where $f(x) = \frac{P(x)}{Q(x)}$ and the degree of $Q(x)$ is greater than the degree of $P(x)$ by at least 2 and $Q(x)$ has no real roots.

To evaluate the integral, consider $\int_C f(z) dz$,

where C is the contour consisting of the upper semicircle C_1 of radius R with centre at the origin and the part of real axis from $-R$ to R (Fig. 5.11).

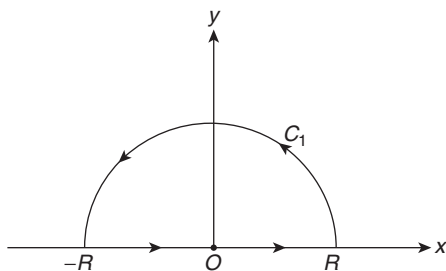


Fig. 5.11

$$\int_C f(z) dz = \int_{C_1} f(z) dz + \int_{-R}^R f(x) dx$$

By Cauchy's residue theorem,

$$\int_C f(z) dz = 2\pi i \text{ (sum of residues at poles in the upper half of the } z\text{-plane)}$$

$$\int_{C_1} f(z) dz + \int_{-R}^R f(x) dx = 2\pi i \text{ (sum of residues)}$$

$$\text{Taking limit } R \rightarrow \infty, \lim_{R \rightarrow \infty} \int_{C_1} f(z) dz + \lim_{R \rightarrow \infty} \int_{-R}^R f(x) dx = 2\pi i \text{ (sum of residues)}$$

$$\text{By Cauchy's lemma, } \lim_{R \rightarrow \infty} \int_{C_1} f(z) dz = 0$$

$$\int_{-\infty}^{\infty} f(x) dx = 2\pi i \text{ (sum of residues)}$$

Example 1

Evaluate $\int_{-\infty}^{\infty} \frac{x^2 dx}{(x^2 + 1)(x^2 + 4)}$ using contour integration.

Solution

$$(i) \text{ Let } \int_C f(z) dz = \int_C \frac{z^2 dz}{(z^2 + 1)(z^2 + 4)}$$

Consider the contour C consisting of the upper semicircle C_1 of radius R and part of the axis from $-R$ to R (Fig. 5.12).

$$(ii) \text{ Let } f(z) = \frac{z^2}{(z^2 + 1)(z^2 + 4)}$$

The poles are given by

$$(z^2 + 1)(z^2 + 4) = 0$$

$$z = \pm i, \pm 2i$$

$$\therefore f(z) = \frac{z^2}{(z+i)(z-i)(z+2i)(z-2i)}$$

(iii) The poles $z = i$ and $z = 2i$ lie inside C .

$$(iv) \text{ Res}[f(z); z = i] = \lim_{z \rightarrow i} [(z - i)f(z)]$$

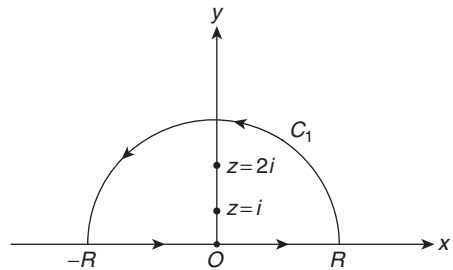


Fig. 5.12

$$\begin{aligned}
&= \lim_{z \rightarrow i} \frac{z^2}{(z+i)(z+2i)(z-2i)} \\
&= \frac{i^2}{(2i)(3i)(-i)} \\
&= -\frac{1}{6i} \\
\text{Res}[f(z); z=2i] &= \lim_{z \rightarrow 2i} [(z-2i)f(z)] \\
&= \lim_{z \rightarrow 2i} \frac{z^2}{(z+i)(z-i)(z+2i)} \\
&= \frac{4i^2}{(3i)(2i)(4i)} \\
&= \frac{-4}{-12i} \\
&= \frac{1}{3i}
\end{aligned}$$

$$(v) \quad \int_{-\infty}^{\infty} f(x) dx = 2\pi i \text{ (sum of residues)}$$

$$\begin{aligned}
\int_{-\infty}^{\infty} \frac{x^2 dx}{(x^2+1)(x^2+4)} &= 2\pi i \left(-\frac{1}{6i} + \frac{1}{3i} \right) \\
&= 2\pi \left(\frac{1}{6} \right) \\
&= \frac{\pi}{3}
\end{aligned}$$

Example 2

Evaluate $\int_{-\infty}^{\infty} \frac{x^2 - x + 2}{x^4 + 10x^2 + 9} dx$ using contour integration.

Solution

$$(i) \quad \text{Let } \int_C f(z) dz = \int_C \frac{z^2 - z + 2}{z^4 + 10z^2 + 9} dz$$

Consider the contour C consisting of the upper semicircle C_1 of radius R and the part of the real axis from $-R$ to R (Fig. 5.13).

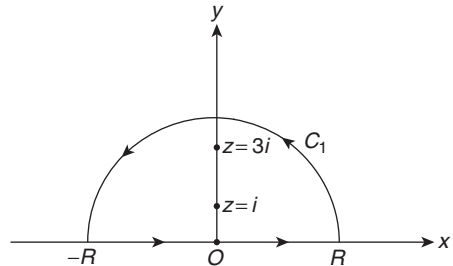


Fig. 5.13

(ii) Let $f(z) = \frac{z^2 - z + 2}{z^4 + 10z^2 + 9}$

The poles are given by

$$z^4 + 10z^2 + 9 = 0$$

$$(z^2 + 1)(z^2 + 9) = 0$$

$$z = \pm i, \pm 3i$$

$$\therefore f(z) = \frac{z^2 - z + 2}{(z+i)(z-i)(z+3i)(z-3i)}$$

(iii) The poles $z = i$ and $z = 3i$ lie inside C .

(iv) $\text{Res}[f(z); z = i] = \lim_{z \rightarrow i} (z - i)f(z)$

$$= \lim_{z \rightarrow i} \frac{z^2 - z + 2}{(z+i)(z+3i)(z-3i)}$$

$$= \frac{i^2 - i + 2}{(2i)(4i)(-2i)}$$

$$= \frac{-1 - i + 2}{16i}$$

$$= \frac{1 - i}{16i}$$

$\text{Res}[f(z); z = 3i] = \lim_{z \rightarrow 3i} (z - 3i)f(z)$

$$= \lim_{z \rightarrow 3i} \frac{z^2 - z + 2}{(z+i)(z-i)(z+3i)}$$

$$= \frac{(3i)^2 - 3i + 2}{(4i)(2i)(6i)}$$

$$= \frac{-9 - 3i + 2}{-48i}$$

$$= \frac{7 + 3i}{48i}$$

(v) $\int_{-\infty}^{\infty} f(x) dx = 2\pi i$ (sum of residues)

$$\int_{-\infty}^{\infty} \frac{x^2 - x + 2}{x^4 + 10x^2 + 9} dx = 2\pi i \left(\frac{1-i}{16i} + \frac{7+3i}{48i} \right)$$

$$= 2\pi i \left[\frac{3(1-i) + 7 + 3i}{48i} \right]$$

$$\begin{aligned}
 &= \frac{10\pi}{24} \\
 &= \frac{5\pi}{12}
 \end{aligned}$$

Example 3

Evaluate $\int_{-\infty}^{\infty} \frac{x^2}{(x^2+1)^2} dx$ using contour integration.

Solution

(i) Let $\int_C f(z) dz = \int_C \frac{z^2}{(z^2+1)^2} dz$

Consider the contour C consisting of the upper semicircle C_1 of radius R and the part of the real axis from $-R$ to R (Fig. 5.14).

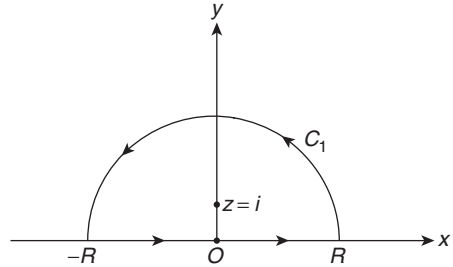


Fig. 5.14

(ii) Let $f(z) = \frac{z^2}{(z^2+1)^2}$

The poles are given by

$$(z^2+1)^2 = 0$$

$$z = \pm i, \pm i$$

$$\therefore f(z) = \frac{z^2}{(z+i)^2(z-i)^2}$$

(iii) The poles $z = i$ of order 2 lies inside C .

$$\begin{aligned}
 \text{(iv) Res}[f(z); z=i] &= \frac{1}{(2-1)!} \lim_{z \rightarrow i} \frac{d}{dz} [(z-i)^2 f(z)] \\
 &= \lim_{z \rightarrow i} \frac{d}{dz} \left[\frac{z^2}{(z+i)^2} \right] \\
 &= \lim_{z \rightarrow i} \frac{(z+i)^2(2z) - z^2 2(z+i)}{(z+i)^4} \\
 &= \lim_{z \rightarrow i} \frac{(z+i)[(z+i)(2z) - 2z^2]}{(z+i)^4} \\
 &= \lim_{z \rightarrow i} \frac{2iz}{(z+i)^3}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{2i \cdot i}{(i+i)^3} \\
 &= \frac{-2}{-8i} \\
 &= \frac{1}{4i}
 \end{aligned}$$

$$(v) \quad \int_{-\infty}^{\infty} f(x) dx = 2\pi i (\text{sum of residues})$$

$$\begin{aligned}
 \int_{-\infty}^{\infty} \frac{x^2}{(x^2+1)^2} dx &= 2\pi i \left(\frac{1}{4i} \right) \\
 &= \frac{\pi}{2}
 \end{aligned}$$

Example 4

Evaluate $\int_0^{\infty} \frac{1}{1+x^2} dx$ using contour integration.

Solution

$$(i) \text{ Let } \int_C f(z) dz = \int_C \frac{1}{z^2+1} dz$$

Consider the contour C consisting of the upper semicircle C_1 of radius R and the part of the real axis from $-R$ to R (Fig. 5.15).

$$(ii) \text{ Let } f(z) = \frac{1}{z^2+1}$$

The poles are given by

$$z^2+1=0$$

$$z = \pm i$$

$$\therefore f(z) = \frac{1}{(z+i)(z-i)}$$

(iii) The pole $z = i$ lies inside C .

$$(iv) \text{ Res}[f(z); z=i] = \lim_{z \rightarrow i} (z-i)f(z)$$

$$\begin{aligned}
 &= \lim_{z \rightarrow i} \frac{1}{z+i} \\
 &= \frac{1}{2i}
 \end{aligned}$$

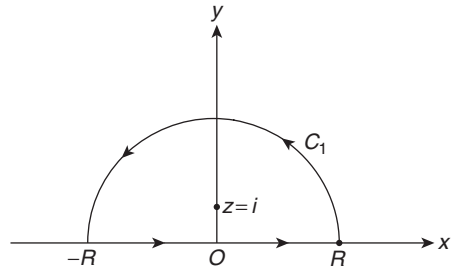


Fig. 5.15

$$\begin{aligned}
 \text{(v)} \quad & \int_{-\infty}^{\infty} f(x) dx = 2\pi i (\text{sum of residues}) \\
 & \int_{-\infty}^{\infty} \frac{1}{1+x^2} dx = 2\pi i \left(\frac{1}{2i} \right) \\
 & \quad = \pi \\
 & 2 \int_0^{\infty} \frac{1}{1+x^2} dx = \pi \quad \left[\because \int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx \text{ if } f(-x) = f(x) \right] \\
 & \int_0^{\infty} \frac{1}{1+x^2} dx = \frac{\pi}{2}
 \end{aligned}$$

Example 5

Evaluate $\int_0^{\infty} \frac{dx}{(1+x^2)^2}$ using contour integration.

Solution

$$\text{(i) Let } \int_C f(z) dz = \int_C \frac{dz}{(1+z^2)^2}$$

Consider the contour C consisting of the upper semicircle C_1 of radius R and the part of the real axis from $-R$ to R (Fig. 5.16).

$$\text{(ii) Let } f(z) = \frac{1}{(1+z^2)^2}$$

The poles are given by

$$(1+z^2)^2 = 0$$

$$1+z^2 = 0$$

$$z^2 = -1$$

$$z = \pm i$$

$$\therefore f(z) = \frac{1}{(z+i)^2(z-i)^2}$$

(iii) The pole $z = i$ of order 2 lies inside C .

$$\text{(iv) } \text{Res}[f(z); z=i] = \frac{1}{(2-1)!} \lim_{z \rightarrow i} \frac{d}{dz} [(z-i)^2 f(z)]$$

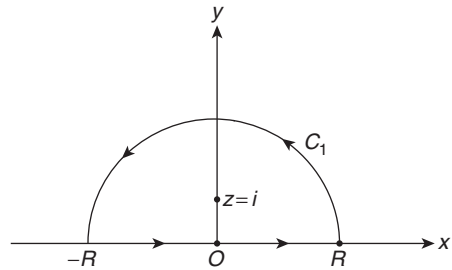


Fig. 5.16

$$\begin{aligned}
 &= \lim_{z \rightarrow i} \frac{d}{dz} \left[\frac{1}{(z+i)^2} \right] \\
 &= \lim_{z \rightarrow i} \left[-\frac{2}{(z+i)^3} \right] \\
 &= -\frac{2}{(i+i)^3} \\
 &= \frac{2}{8i} \\
 &= \frac{1}{4i}
 \end{aligned}$$

$$(v) \quad \int_{-\infty}^{\infty} f(x) dx = 2\pi i (\text{sum of residues})$$

$$\begin{aligned}
 \int_{-\infty}^{\infty} \frac{1}{(1+x^2)^2} dx &= 2\pi i \left(\frac{1}{4i} \right) \\
 &= \frac{\pi}{2}
 \end{aligned}$$

$$2 \int_0^{\infty} \frac{1}{(1+x^2)^2} dx = \frac{\pi}{2} \quad \left[\because \int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx \text{ if } f(-x) = f(x) \right]$$

$$\int_0^{\infty} \frac{1}{(1+x^2)^2} dx = \frac{\pi}{4}$$

Example 6

Evaluate $\int_0^{\infty} \frac{x^2 dx}{(x^2 + a^2)(x^2 + b^2)}$, $a > 0$, $b > 0$

Solution

$$(i) \text{ Let } \int_C f(z) dz = \int_C \frac{z^2 dz}{(z^2 + a^2)(z^2 + b^2)}$$

Consider the contour C consisting of the upper semicircle C_1 of radius R and the part of the real axis from $-R$ to R (Fig. 5.17).

$$(ii) \text{ Let } f(z) = \frac{z^2}{(z^2 + a^2)(z^2 + b^2)}$$

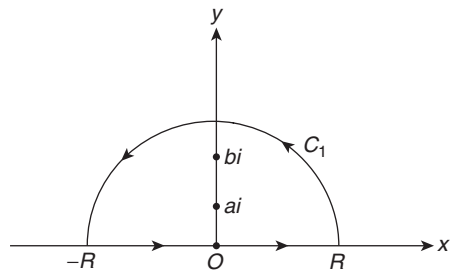


Fig. 5.17

The poles are given by

$$(z^2 + a^2)(z^2 + b^2) = 0$$

$$z = \pm ai, \pm bi$$

$$\therefore f(z) = \frac{z^2}{(z+ai)(z-ai)(z+bi)(z-bi)}$$

(iii) The poles $z = ai$ and $z = bi$ lie inside C as $a > 0, b > 0$.

$$(iv) \quad \text{Res}[f(z); z = ai] = \lim_{z \rightarrow ai} (z - ai)f(z)$$

$$\begin{aligned} &= \lim_{z \rightarrow ai} \frac{z^2}{(z+ai)(z+bi)(z-bi)} \\ &= \frac{a^2 i^2}{(2ai)(ai+bi)(ai-bi)} \\ &= -\frac{a}{2i(b^2 - a^2)} \end{aligned}$$

$$\text{Res}[f(z); z = bi] = \lim_{z \rightarrow bi} (z - bi)f(z)$$

$$\begin{aligned} &= \lim_{z \rightarrow bi} \frac{z^2}{(z+ai)(z-ai)(z+bi)} \\ &= \frac{b^2 i^2}{(bi+ai)(bi-ai)(2bi)} \\ &= -\frac{b}{2i(a^2 - b^2)} \\ &= \frac{b}{2i(b^2 - a^2)} \end{aligned}$$

$$(v) \quad \int_{-\infty}^{\infty} f(x) dx = 2\pi i (\text{sum of residues})$$

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{x^2}{(x^2 + a^2)(x^2 + b^2)} dx &= 2\pi i \left[-\frac{a}{2i(b^2 - a^2)} + \frac{b}{2i(b^2 - a^2)} \right] \\ &= \pi \left[\frac{b-a}{b^2 - a^2} \right] \\ &= \frac{\pi}{b+a} \end{aligned}$$

$$2 \int_0^{\infty} \frac{x^2}{(x^2 + a^2)(x^2 + b^2)} dx = \frac{\pi}{b+a} \left[\because \int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx \text{ if } f(-x) = f(x) \right]$$

$$\int_0^{\infty} \frac{x^2}{(x^2 + a^2)(x^2 + b^2)} dx = \frac{\pi}{2(b+a)}$$

Example 7

Show that $\int_0^{\infty} \frac{dx}{1+x^4} = \frac{\pi}{2\sqrt{2}}.$

Solution

(i) Let $\int_C f(z) dz = \int_C \frac{dz}{1+z^4}$

Consider the contour C consisting of the upper semicircle C_1 of radius R and the part of the real axis from $-R$ to R (Fig. 5.18).

(ii) Let $f(z) = \frac{1}{1+z^4}$

The poles are given by

$$1+z^4=0$$

$$z^4 = -1$$

$$z = (-1)^{\frac{1}{4}}$$

$$= e^{\frac{i\pi}{4}}, e^{\frac{-i\pi}{4}}, e^{\frac{i3\pi}{4}}, e^{\frac{-i3\pi}{4}}$$

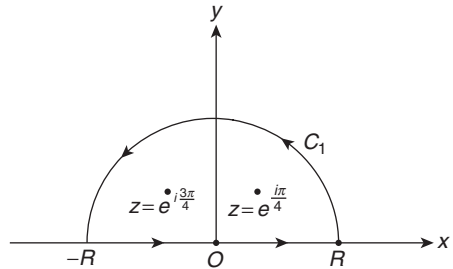


Fig. 5.18

(iii) The poles $z = e^{\frac{i\pi}{4}}$ and $z = e^{\frac{i3\pi}{4}}$ lie inside C .

(iv) $\text{Res}[f(z); z = e^{\frac{i\pi}{4}}] = \lim_{z \rightarrow e^{\frac{i\pi}{4}}} (z - e^{\frac{i\pi}{4}}) f(z)$

$$= \lim_{z \rightarrow e^{\frac{i\pi}{4}}} \frac{z - e^{\frac{i\pi}{4}}}{1 + z^4} \quad \left[\frac{0}{0} \text{ form} \right]$$

$$= \lim_{z \rightarrow e^{\frac{i\pi}{4}}} \frac{1}{4z^3}$$

$$= \frac{1}{4e^{\frac{3i\pi}{4}}}$$

$$= \frac{1}{4} e^{-\frac{3i\pi}{4}}$$

$$= \frac{1}{4} \left(\cos \frac{3\pi}{4} - i \sin \frac{3\pi}{4} \right)$$

$$= \frac{1}{4} \left(-\frac{1}{\sqrt{2}} - i - \frac{1}{\sqrt{2}} \right)$$

$$\begin{aligned}
&= -\frac{(1+i)}{4\sqrt{2}} \\
\text{Res}[f(z); z = e^{i\frac{3\pi}{4}}] &= \lim_{z \rightarrow e^{i\frac{3\pi}{4}}} (z - e^{i\frac{3\pi}{4}}) f(z) \\
&= \lim_{z \rightarrow e^{i\frac{3\pi}{4}}} \frac{z - e^{i\frac{3\pi}{4}}}{1 + z^4} \quad \left[\frac{0}{0} \text{ form} \right] \\
&= \lim_{z \rightarrow e^{i\frac{3\pi}{4}}} \frac{1}{4z^3} \quad [\text{Applying L'Hospital rule}] \\
&= \frac{1}{4e^{i\frac{9\pi}{4}}} \\
&= \frac{1}{4} e^{-i\frac{9\pi}{4}} \\
&= \frac{1}{4} \left(\cos \frac{9\pi}{4} - i \sin \frac{9\pi}{4} \right) \\
&= \frac{1}{4} \left(\frac{1}{\sqrt{2}} - i \frac{1}{\sqrt{2}} \right) \\
&= \frac{1-i}{4\sqrt{2}}
\end{aligned}$$

$$(v) \quad \int_{-\infty}^{\infty} f(x) dx = 2\pi i (\text{sum of residues})$$

$$\begin{aligned}
\int_{-\infty}^{\infty} \frac{1}{1+x^4} dx &= 2\pi i \left[-\frac{1+i}{4\sqrt{2}} + \frac{1-i}{4\sqrt{2}} \right] \\
&= 2\pi i \left(-\frac{2i}{4\sqrt{2}} \right) \\
&= \frac{\pi}{\sqrt{2}}
\end{aligned}$$

$$2 \int_0^{\infty} \frac{1}{1+x^4} dx = \frac{\pi}{\sqrt{2}} \quad \left[\because \int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx \text{ if } f(-x) = f(x) \right]$$

$$\int_0^{\infty} \frac{1}{1+x^4} dx = \frac{\pi}{2\sqrt{2}}$$

Example 8

Using the theory of residues, evaluate $\int_{-\infty}^{\infty} \frac{\cos x}{x^2 + 1} dx$. [Summer 2015]

Solution

(i) Let $\int_C f(z) dz = \int_C \frac{e^{iz}}{z^2 + 1} dz$

Consider the contour C consisting of the upper semicircle C_1 of radius R and the part of the real axis from $-R$ to R (Fig. 5.19).

(ii) Let $f(z) = \frac{e^{iz}}{z^2 + 1}$

The poles are given by

$$z^2 + 1 = 0$$

$$z = \pm i$$

$$\therefore f(z) = \frac{e^{iz}}{(z+i)(z-i)}$$

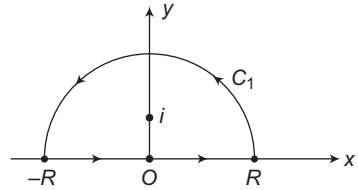


Fig. 5.19

(iii) The pole $z = i$ lies inside C .

(iv) $\text{Res}[f(z); z = i] = \lim_{z \rightarrow i} (z - i) f(z)$

$$= \lim_{z \rightarrow i} \frac{e^{iz}}{(z + i)}$$

$$= \frac{e^{i^2}}{2i}$$

$$= \frac{e^{-1}}{2i}$$

$$= \frac{1}{2ie}$$

(v) $\int_{-\infty}^{\infty} f(x) dx = 2\pi i$ (sum of residues)

$$\int_{-\infty}^{\infty} \frac{e^{ix}}{x^2 + 1} = 2\pi i \left(\frac{1}{2ie} \right)$$

$$\int_{-\infty}^{\infty} \frac{\cos x + i \sin x}{x^2 + 1} dx = \frac{\pi}{e}$$

Comparing real parts,

$$\int_{-\infty}^{\infty} \frac{\cos x}{x^2 + 1} dx = \frac{\pi}{e}$$

EXERCISE 5.2

Evaluate the following integrals using contour integration:

$$1. \int_0^{\infty} \frac{x^2 dx}{(x^2 + 9)(x^2 + 4)} \quad \left[\text{Ans.: } \frac{\pi}{10} \right]$$

$$2. \int_0^{\infty} \frac{dx}{(x^2 + a^2)(x^2 + b^2)}, a > b > 0 \quad \left[\text{Ans.: } \frac{\pi}{2ab(a+b)} \right]$$

$$3. \int_0^{\infty} \frac{dx}{x^4 + 10x^2 + 9} \quad \left[\text{Ans.: } \frac{\pi}{24} \right]$$

$$4. \int_{-\infty}^{\infty} \frac{dx}{(x^2 + 1)^3} \quad \left[\text{Ans.: } \frac{3\pi}{8} \right]$$

$$5. \int_0^{\infty} \frac{dx}{(x^2 + a^2)^2}, a > 0 \quad \left[\text{Ans.: } \frac{\pi}{4a^3} \right]$$

$$6. \int_{-\infty}^{\infty} \frac{x^2 + x + 3}{x^4 + 5x^2 + 4} dx \quad \left[\text{Ans.: } \frac{5\pi}{6} \right]$$

$$7. \int_0^{\infty} \frac{x^2}{(x^2 + 1)^3} dx \quad \left[\text{Ans.: } \frac{\pi}{6} \right]$$

$$8. \int_{-\infty}^{\infty} \frac{x^2}{(x^2 + a^2)^2} dx, a > 0 \quad \left[\text{Ans.: } \frac{3\pi}{2a} \right]$$

$$9. \int_{-\infty}^{\infty} \frac{x^2}{(x^2 + 1)(x^2 + 2x + 2)} dx \quad \left[\text{Ans.: } \frac{3\pi}{5} \right]$$

$$10. \int_{-\infty}^{\infty} \frac{x^2 dx}{(x^2 + 9)(x^2 + 25)} \quad \left[\text{Ans.: } \frac{\pi}{8} \right]$$

5.4 EVALUATION OF IMPROPER REAL INTEGRAL OF A RATIONAL FUNCTION INCLUDING TRIGONOMETRIC FUNCTIONS

$$\text{Let } I = \int_{-\infty}^{\infty} \frac{\cos mx}{Q(x)} dx \quad \text{or} \quad \int_{-\infty}^{\infty} \frac{\sin mx}{Q(x)} dx$$

where $Q(x)$ is a polynomial in x .

To evaluate the integral,

$$\text{consider } \int_C f(z) dz = \int e^{imz} g(z) dz$$

where $g(z) = \frac{1}{Q(z)}$ and C is the contour

consisting of the upper semicircle C_1 of radius R with centre at the origin and the part of the real axis from $-R$ to R (Fig. 5.20).

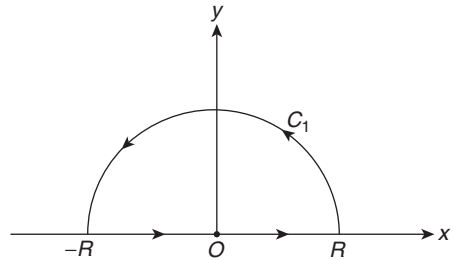


Fig. 5.20

$$\int_C f(z) dz = \int_{-R}^R f(x) dx + \int_{C_1} f(z) dz$$

By Cauchy's residue theorem,

$$\int_C f(z) dz = 2\pi i \quad (\text{sum of residues at poles in the upper half of } z\text{-plane})$$

$$\int_{-R}^R f(x) dx + \int_{C_1} f(z) dz = 2\pi i \quad (\text{sum of residues})$$

Taking $R \rightarrow \infty$,

$$\lim_{R \rightarrow \infty} \int_{-R}^R f(x) dx + \lim_{R \rightarrow \infty} \int_{C_1} f(z) dz = 2\pi i (\text{sum of residues})$$

$$\int_{-\infty}^{\infty} f(x) dx + \lim_{R \rightarrow \infty} \int_{C_1} e^{imz} g(z) dz = 2\pi i (\text{sum of residues})$$

By Jordan's lemma, $\lim_{R \rightarrow \infty} \int_{C_1} e^{imz} g(z) dz = 0$

$$\int_{-\infty}^{\infty} f(x) dx = 2\pi i (\text{sum of residues})$$

Example 1

Evaluate $\int_0^{\infty} \frac{\cos mx}{x^2 + a^2} dx$, $a > 0, m > 0$ using contour integration.

Solution

(i) Let $\int_C f(z) dz = \int_C \frac{e^{imz}}{z^2 + a^2} dz$

Consider the contour C consisting of the upper semicircle C_1 of radius R and the part of the real axis from $-R$ to R (Fig. 5.21).

(ii) Let $f(z) = \frac{e^{imz}}{z^2 + a^2}$

The poles are given by

$$z^2 + a^2 = 0$$

$$z = \pm ai$$

$$\therefore f(z) = \frac{e^{imz}}{(z + ai)(z - ai)}$$

(iii) The pole $z = ai$ lies inside C .

(iv) $\text{Res}[f(z); z = ai] = \lim_{z \rightarrow ai} (z - ai)f(z)$

$$= \lim_{z \rightarrow ai} \frac{e^{imz}}{(z + ai)}$$

$$= \frac{e^{im(ai)}}{(ai + ai)}$$

$$= \frac{e^{-ma}}{2ai}$$

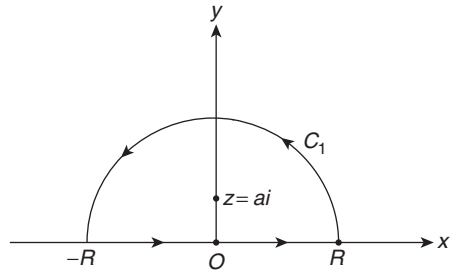


Fig. 5.21

$$(v) \quad \int_{-\infty}^{\infty} f(x) dx = 2\pi i \text{ (sum of residues)}$$

$$\int_{-\infty}^{\infty} \frac{e^{imx}}{x^2 + a^2} dx = 2\pi i \left(\frac{e^{-ma}}{2ai} \right)$$

$$= \frac{\pi e^{-ma}}{a}$$

$$\int_{-\infty}^{\infty} \frac{(\cos mx + i \sin mx)}{x^2 + a^2} dx = \frac{\pi e^{-ma}}{a}$$

Comparing real parts,

$$\int_{-\infty}^{\infty} \frac{\cos mx}{x^2 + a^2} dx = \frac{\pi e^{-ma}}{a}$$

$$2 \int_0^{\infty} \frac{\cos mx}{x^2 + a^2} dx = \frac{\pi e^{-ma}}{a} \quad \left[\because \int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx \text{ if } f(-x) = f(x) \right]$$

$$\int_0^{\infty} \frac{\cos mx}{x^2 + a^2} dx = \frac{\pi e^{-ma}}{2a}$$

Example 2

Evaluate a real integral $\int_0^{\infty} \frac{x \sin x}{x^2 + 9} dx$ using residue. [Winter 2013]

Solution

$$(i) \text{ Let } \int_C f(z) dz = \int_C \frac{ze^{iz}}{z^2 + 9} dz$$

Consider the contour C consisting of the upper semicircle C_1 of radius R and the part of the real axis from $-R$ to R (Fig. 5.22).

$$(ii) \text{ Let } f(z) = \frac{ze^{iz}}{z^2 + 9}$$

The poles are given by

$$z^2 + 9 = 0$$

$$z = \pm 3i$$

$$\therefore f(z) = \frac{ze^{iz}}{(z + 3i)(z - 3i)}$$

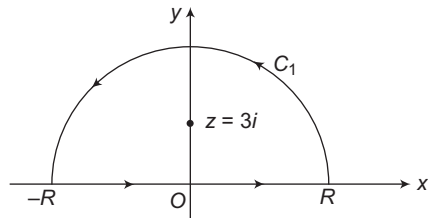


Fig. 5.22

(iii) The pole $z = 3i$ lies inside C .

$$(iv) \operatorname{Res}[f(z); z = 3i] = \lim_{z \rightarrow 3i} (z - 3i) f(z)$$

$$= \lim_{z \rightarrow 3i} \frac{ze^{iz}}{z + 3i}$$

$$= \frac{3ie^{3i^2}}{3i + 3i}$$

$$= \frac{3ie^{-3}}{6i}$$

$$= \frac{e^{-3}}{2}$$

$$= \frac{1}{2e^3}$$

$$(v) \int_{-\infty}^{\infty} f(x) dx = 2\pi i (\text{sum of residues})$$

$$\int_{-\infty}^{\infty} \frac{xe^{ix}}{x^2 + 9} dx = 2\pi i \left(\frac{1}{2e^3} \right)$$

$$\int_{-\infty}^{\infty} \frac{x(\cos x + i \sin x)}{x^2 + 9} dx = \frac{\pi i}{e^3}$$

Comparing imaginary parts,

$$\int_{-\infty}^{\infty} \frac{x \sin x}{x^2 + 9} dx = \frac{\pi}{e^3}$$

$$2 \int_0^{\infty} \frac{x \sin x}{x^2 + 9} dx = \frac{\pi}{e^3} \quad \left[\because \int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx, \text{ if } f(-x) = f(x) \right]$$

$$\int_0^{\infty} \frac{x \sin x}{x^2 + 9} dx = \frac{\pi}{2e^3}$$

Example 3

Evaluate $\int_0^{\infty} \frac{x \sin mx}{x^2 + a^2} dx, a > 0, m > 0$.

Solution

$$(i) \text{ Let } \int_C f(z) dz = \frac{ze^{imz}}{z^2 + a^2} dz$$

Consider the contour C consisting of upper semicircle C_1 of radius R and the part of the real axis from $-R$ to R (Fig. 5.23).

$$(ii) \text{ Let } f(z) = \frac{ze^{imz}}{z^2 + a^2}$$

The poles are given by

$$z^2 + a^2 = 0$$

$$z = \pm ai$$

$$\therefore f(z) = \frac{ze^{imz}}{(z+ai)(z-ai)}$$

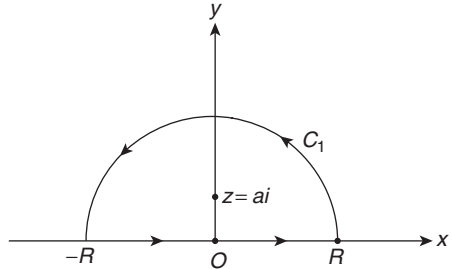


Fig. 5.23

(iii) The pole $z = ai$ lies inside C .

$$(iv) \text{ Res}[f(z); z = ai] = \lim_{z \rightarrow ai} (z - ai)f(z)$$

$$= \lim_{z \rightarrow ai} \frac{ze^{imz}}{z + ai}$$

$$= \frac{aie^{i^2ma}}{ai + ai}$$

$$= \frac{aie^{-ma}}{2ai}$$

$$= \frac{e^{-ma}}{2}$$

$$(v) \int_{-\infty}^{\infty} f(x) dx = 2\pi i (\text{sum of residues})$$

$$\int_{-\infty}^{\infty} \frac{xe^{imx}}{x^2 + a^2} dx = 2\pi i \left(\frac{e^{-ma}}{2} \right)$$

$$= \pi i e^{-ma}$$

$$\int_{-\infty}^{\infty} \frac{x(\cos mx + i \sin mx)}{x^2 + a^2} dx = \pi i e^{-ma}$$

Comparing imaginary parts,

$$\int_{-\infty}^{\infty} \frac{x \sin mx}{x^2 + a^2} dx = \pi e^{-ma}$$

$$2 \int_0^{\infty} \frac{x \sin mx}{x^2 + a^2} dx = \pi e^{-ma} \left[\because \int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx \text{ if } f(-x) = f(x) \right]$$

$$\int_0^{\infty} \frac{x \sin mx}{x^2 + a^2} dx = \frac{\pi e^{-ma}}{2}$$

Example 4

Evaluate $\int_0^{\infty} \frac{\cos mx}{(1+x^2)^2} dx, m > 0$ using contour integration.

Solution

(i) Let $f(z)dz = \int_C \frac{e^{imz}}{(1+z^2)^2} dz$

Consider the contour C consisting of the upper semicircle of radius R and the part of the real axis from $-R$ to R (Fig. 5.24).

(ii) Let $f(z) = \frac{e^{imz}}{(1+z^2)^2}$

The poles are given by

$$(1+z^2)^2 = 0$$

$$1+z^2 = 0$$

$$z = \pm i$$

$$\therefore f(z) = \frac{e^{imz}}{(z+i)^2(z-i)^2}$$

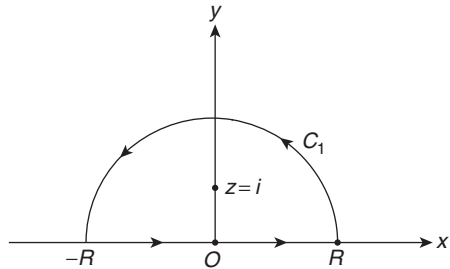


Fig. 5.24

(iii) The pole $z = i$ of order 2 lies inside C .

$$\begin{aligned} \text{(iv) } \text{Res}[f(z); z = i] &= \frac{1}{(2-1)!} \lim_{z \rightarrow i} \frac{d}{dz} [(z-i)^2 f(z)] \\ &= \lim_{z \rightarrow i} \frac{d}{dz} \left[\frac{e^{imz}}{(z+i)^2} \right] \\ &= \lim_{z \rightarrow i} \frac{(z+i)^2 e^{imz} \cdot im - e^{imz} 2(z+i)}{(z+i)^4} \\ &= \lim_{z \rightarrow i} \frac{[im(z+i) - 2]e^{imz}}{(z+i)^3} \\ &= \frac{[(i+i)im - 2]e^{i^2 m}}{(i+i)^3} \\ &= \frac{(-2m-2)e^{-m}}{-8i} \\ &= \frac{(m+1)e^{-m}}{4i} \end{aligned}$$

$$(v) \quad \int_{-\infty}^{\infty} f(x) dx = 2\pi i (\text{sum of residues})$$

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{e^{imx}}{(1+x^2)^2} dx &= 2\pi i \left[\frac{(m+1)e^{-m}}{4i} \right] \\ &= \frac{\pi(m+1)e^{-m}}{2} \end{aligned}$$

$$\int_{-\infty}^{\infty} \frac{\cos mx + i \sin mx}{(1+x^2)^2} dx = \frac{\pi(m+1)e^{-m}}{2}$$

Comparing real parts,

$$\int_{-\infty}^{\infty} \frac{\cos mx}{(1+x^2)^2} dx = \frac{\pi(m+1)e^{-m}}{2}$$

$$2 \int_0^{\infty} \frac{\cos mx}{(1+x^2)^2} dx = \frac{\pi(m+1)e^{-m}}{2}$$

$$\int_0^{\infty} \frac{\cos mx}{(1+x^2)^2} dx = \frac{\pi(m+1)e^{-m}}{4}$$

$$\left[\because \int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx \text{ if } f(-x) = f(x) \right]$$

Example 5

Evaluate $\int_{-\infty}^{\infty} \frac{\cos x dx}{(x^2 + a^2)(x^2 + b^2)}$, $a > 0, b > 0$ using contour integration.

Solution

$$(i) \text{ Let } \int_C f(z) dz = \int_C \frac{e^{iz} dz}{(z^2 + a^2)(z^2 + b^2)}$$

Consider the contour C consisting of the upper semicircle C_1 of radius R and the part of the real axis from $-R$ to R (Fig. 5.25).

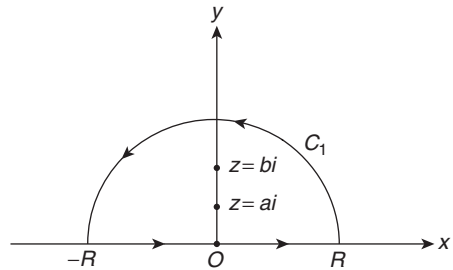


Fig. 5.25

$$(ii) \text{ Let } f(z) = \frac{1}{(z^2 + a^2)(z^2 + b^2)}$$

The poles are given by

$$(z^2 + a^2)(z^2 + b^2) = 0$$

$$z = \pm ai, \pm bi$$

$$\therefore f(z) = \frac{1}{(z+ai)(z-ai)(z+bi)(z-bi)}$$

(iii) The poles $z = ai$ and $z = bi$ lie inside C .

$$\begin{aligned} \text{(iv) } \operatorname{Res}[f(z); z = ai] &= \lim_{z \rightarrow ai} (z - ai)f(z) \\ &= \lim_{z \rightarrow ai} \frac{e^{iz}}{(z+ai)(z+bi)(z-bi)} \\ &= \frac{e^{i(ai)}}{(2ai)(ai+bi)(ai-bi)} \\ &= \frac{e^{-a}}{-2ai(a^2 - b^2)} \end{aligned}$$

$$\begin{aligned} \operatorname{Res}[f(z); z = bi] &= \lim_{z \rightarrow bi} (z - bi)f(z) \\ &= \lim_{z \rightarrow bi} \frac{e^{iz}}{(z+ai)(z-ai)(z+bi)} \\ &= \frac{e^{i(bi)}}{(bi+ai)(bi-ai)(2bi)} \\ &= \frac{e^{-b}}{2bi(a^2 - b^2)} \end{aligned}$$

$$\text{(v) } \int_{-\infty}^{\infty} f(x) dx = 2\pi i \text{ (sum of residues)}$$

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{e^{ix}}{(x^2+a^2)(x^2+b^2)} dx &= 2\pi i \left[\frac{e^{-a}}{-2ai(a^2-b^2)} + \frac{e^{-b}}{2bi(a^2-b^2)} \right] \\ &= \frac{\pi}{a^2-b^2} \left(\frac{e^{-b}}{b} - \frac{e^{-a}}{a} \right) \end{aligned}$$

$$\int_{-\infty}^{\infty} \frac{\cos x + i \sin x}{(x^2+a^2)(x^2+b^2)} dx = \frac{\pi}{a^2-b^2} \left(\frac{e^{-b}}{b} - \frac{e^{-a}}{a} \right)$$

Comparing real parts,

$$\int_{-\infty}^{\infty} \frac{\cos x dx}{(x^2+a^2)(x^2+b^2)} = \frac{\pi}{a^2-b^2} \left(\frac{e^{-b}}{b} - \frac{e^{-a}}{a} \right)$$

Example 6

Evaluate $\int_0^{\infty} \frac{x^2 \cos mx}{(x^2 + a^2)(x^2 + b^2)} dx, m > 0, a > b > 0.$

Solution

(i) Let $\int_C f(z) dz = \int_C \frac{z^2 e^{imz}}{(z^2 + a^2)(z^2 + b^2)} dz$

Consider the contour C consisting of the upper semicircle C_1 of radius R and the part of the real axis from $-R$ to R (Fig. 5.26).

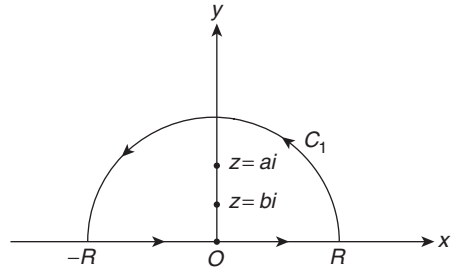


Fig. 5.26

(ii) Let $f(z) = \frac{z^2 e^{imz}}{(z^2 + a^2)(z^2 + b^2)}$

The poles are given by

$$(z^2 + a^2)(z^2 + b^2) = 0$$

$$z = \pm ai, \pm bi$$

$$\therefore f(z) = \frac{z^2 e^{imz}}{(z + ai)(z - ai)(z + bi)(z - bi)}$$

(iii) The poles $z = ai$ and $z = bi$ lie inside C .

(iv) $\text{Res}[f(z); z = ai] = \lim_{z \rightarrow ai} (z - ai)f(z)$

$$= \lim_{z \rightarrow ai} \frac{z^2 e^{imz}}{(z + ai)(z + bi)(z - bi)}$$

$$= \frac{(ai)^2 e^{i^2 ma}}{(2ai)(ai + bi)(ai - bi)}$$

$$= \frac{-a^2 e^{-ma}}{2ai(b^2 - a^2)}$$

$$= \frac{-ae^{-ma}}{2i(b^2 - a^2)}$$

$\text{Res}[f(z); z = bi] = \lim_{z \rightarrow bi} (z - bi)f(z)$

$$= \lim_{z \rightarrow bi} \frac{z^2 e^{imz}}{(z + ai)(z - ai)(z + bi)}$$

$$\begin{aligned}
 &= \frac{(bi)^2 e^{i^2 mb}}{(bi+ai)(bi-ai)(2bi)} \\
 &= \frac{-b^2 e^{-mb}}{(a^2 - b^2)(2bi)} \\
 &= \frac{be^{-mb}}{2i(b^2 - a^2)}
 \end{aligned}$$

$$(v) \quad \int_{-\infty}^{\infty} f(x) dx = 2\pi i \text{ (sum of residues)}$$

$$\begin{aligned}
 \int_{-\infty}^{\infty} \frac{x^2 e^{imx}}{(x^2 + a^2)(x^2 + b^2)} dx &= 2\pi i \left[-\frac{ae^{-ma}}{2i(b^2 - a^2)} + \frac{be^{-mb}}{2i(b^2 - a^2)} \right] \\
 &= \pi \left[\frac{be^{-mb} - ae^{-ma}}{b^2 - a^2} \right]
 \end{aligned}$$

$$\int_{-\infty}^{\infty} \frac{x^2 (\cos mx + i \sin mx)}{(x^2 + a^2)(x^2 + b^2)} dx = \frac{\pi(be^{-mb} - ae^{-ma})}{b^2 - a^2}$$

Comparing real parts,

$$\int_{-\infty}^{\infty} \frac{x^2 \cos mx}{(x^2 + a^2)(x^2 + b^2)} dx = \frac{\pi(be^{-mb} - ae^{-ma})}{(b^2 - a^2)}$$

$$2 \int_0^{\infty} \frac{x^2 \cos mx}{(x^2 + a^2)(x^2 + b^2)} dx = \frac{\pi(be^{-mb} - ae^{-ma})}{(b^2 - a^2)} \quad \left[\begin{array}{l} \because \int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx \\ \text{if } f(-x) = f(x) \end{array} \right]$$

$$\int_0^{\infty} \frac{x^2 \cos mx}{(x^2 + a^2)(x^2 + b^2)} dx = \frac{\pi(be^{-mb} - ae^{-ma})}{2(b^2 - a^2)}$$

EXERCISE 5.3

Evaluate the following integrals using contour integration:

$$1. \int_{-\infty}^{\infty} \frac{\cos mx}{(x^2 + a^2)^2} dx, a > 0, m > 0$$

$$\left[\text{Ans.: } \frac{\pi}{4a^3} (1 + ma) e^{-ma} \right]$$

$$2. \int_{-\infty}^{\infty} \frac{\cos x}{1 + x^2} dx$$

$$\left[\text{Ans.: } \frac{\pi}{e} \right]$$

$$3. \int_{-\infty}^{\infty} \frac{x \sin x}{x^2 + 2x + 2} dx \quad \left[\text{Ans.: } \frac{\pi}{e} (\sin 1 + 1) \right]$$

$$4. \int_{-\infty}^{\infty} \frac{\cos x}{x^4 + 5x^2 + 4} dx \quad \left[\text{Ans.: } \pi \left(\frac{1}{3e} - \frac{1}{6e^2} \right) \right]$$

$$5. \int_{-\infty}^{\infty} \frac{\sin x}{x^2 + 4x + 5} dx \quad \left[\text{Ans.: } -\frac{\pi}{e} \sin 2 \right]$$

$$6. \int_{-\infty}^{\infty} \frac{\cos ax}{x^4 + 10x^2 + 9} dx, a > 0 \quad \left[\text{Ans.: } \frac{\pi}{8} \left(e^{-a} - \frac{e^{-3a}}{3} \right) \right]$$

5.5 EVALUATION OF IMPROPER INTEGRAL WHEN SIMPLE POLES LIE ON THE REAL AXIS

$$\text{Let } I = \int_{-\infty}^{\infty} f(x) dx$$

To evaluate the integral, consider $\int_C f(z) dz$.

If $f(z)$ has a simple pole on the real axis then this pole is deleted by indenting the contour. The contour is indented by drawing a small semi-circle C_1 of radius r having the pole as its centre (Fig. 5.27).

If $f(z)$ has a simple pole at $z = b$ on the real axis then

$$\lim_{r \rightarrow 0} \int_{C_1} f(z) dz = \pi i \operatorname{Res} [f(z); z = b]$$

If $f(z)$ has a simple pole $z = b$ on the real axis and another simple pole $z = z_1$ in the upper half of the z -plane then

$$\int_{-\infty}^{\infty} f(x) dx = \pi i \operatorname{Res} [f(z); z = b] + 2\pi i \operatorname{Res} [f(z); z = z_1]$$

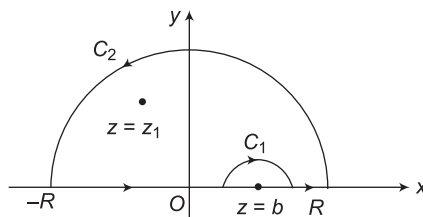


Fig. 5.27

Example 1

Find the principal value of $\int_{-\infty}^{\infty} \frac{dx}{(x^2 - 3x + 2)(x^2 + 1)}$.

Solution

(i) Let $\int_C f(z) dz = \int_C \frac{dz}{(z^2 - 3z + 2)(z^2 + 1)}$

(ii) Let $f(z) = \frac{1}{(z^2 - 3z + 2)(z^2 + 1)}$

The poles are given by

$$(z^2 - 3z + 2)(z^2 + 1) = 0$$

$$(z - 1)(z - 2)(z + i)(z - i) = 0$$

$$z = 1, 2, \pm i$$

$$\therefore f(z) = \frac{1}{(z - 1)(z - 2)(z + i)(z - i)}$$

(iii) The poles $z = 1, z = 2$ lie on the real axis and the pole $z = i$ lies in the upper half of the z -plane (Fig. 5.28).

(iv) $\text{Res}[f(z); z = 1] = \lim_{z \rightarrow 1} (z - 1)f(z)$

$$= \lim_{z \rightarrow 1} \frac{1}{(z - 2)(z^2 + 1)}$$

$$= -\frac{1}{2}$$

$$\text{Res}[f(z); z = 2] = \lim_{z \rightarrow 2} (z - 2)f(z)$$

$$= \lim_{z \rightarrow 2} \frac{1}{(z - 1)(z^2 + 1)}$$

$$= \frac{1}{5}$$

$$\text{Res}[f(z); z = i] = \lim_{z \rightarrow i} (z - i)f(z)$$

$$= \lim_{z \rightarrow i} \frac{1}{(z^2 - 3z + 2)(z + i)}$$

$$= \frac{1}{(-1 - 3i + 2)2i}$$

$$= \frac{1}{(1 - 3i)2i}$$

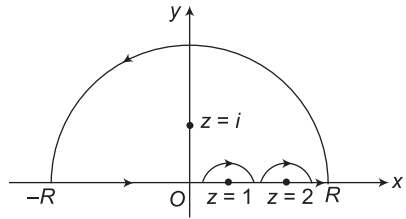


Fig. 5.28

$$\begin{aligned}
 &= \frac{1}{2(i+3)} \times \frac{(-i+3)}{(-i+3)} \\
 &= \frac{-i+3}{2(1+9)} \\
 &= \frac{3-i}{20}
 \end{aligned}$$

$$\begin{aligned}
 \text{(v) PV of } \int_{-\infty}^{\infty} \frac{dx}{(x^2-3x+2)(x^2+1)} &= \pi i \left\{ \text{Res} [f(z); z=1] + [f(z); z=2] \right\} \\
 &\quad + 2\pi i \text{Res} [f(z); z=i] \\
 &= \pi i \left(-\frac{1}{2} + \frac{1}{5} \right) + 2\pi i \left(\frac{3-i}{20} \right) \\
 &= \pi i \left(-\frac{3}{10} \right) + \pi i \left(\frac{3-i}{10} \right) \\
 &= -\frac{3\pi i}{10} + \frac{3\pi i}{10} - \frac{\pi i^2}{10} \\
 &= \frac{\pi}{10}
 \end{aligned}$$

Example 2

Show that the principal value of $\int_{-\infty}^{\infty} \frac{\sin x \, dx}{(x^2+4)(x-1)} = \frac{\pi}{5} \left(\cos 1 - \frac{1}{e^2} \right)$.

Solution

$$\text{(i) Let } \int_C f(z) \, dz = \int_C \frac{e^{iz}}{(z^2+4)(z-1)} \, dz$$

$$\text{(ii) Let } f(z) = \frac{e^{iz}}{(z^2+4)(z-1)}$$

The poles are given by

$$(z^2+4)(z-1) = 0$$

$$z = \pm 2i, 1$$

$$\therefore f(z) = \frac{e^{iz}}{(z+2i)(z-2i)(z-1)}$$

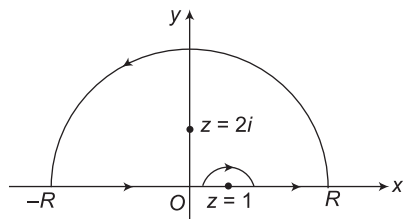


Fig. 5.29

(iii) The pole $z = 1$ lies on the real axis and the pole $z = 2i$ lies in the upper half of the z -plane (Fig. 5.29).

$$(iv) \operatorname{Res} [f(z); z = 1] = \lim_{z \rightarrow 1} (z - 1)f(z)$$

$$= \lim_{z \rightarrow 1} \frac{e^{iz}}{(z^2 + 4)}$$

$$= \frac{e^i}{5}$$

$$\operatorname{Res} [f(z); z = 2i] = \lim_{z \rightarrow 2i} (z - 2i) f(z)$$

$$= \lim_{z \rightarrow 2i} \frac{e^{iz}}{(z + 2i)(z - 1)}$$

$$= \frac{e^{i(2i)}}{4i(2i - 1)}$$

$$= \frac{e^{-2}}{-8 - 4i}$$

$$= \frac{e^{-2}}{-4(2 + i)} \times \frac{(2 - i)}{(2 - i)}$$

$$= \frac{e^{-2}(2 - i)}{-4(5)}$$

$$= \frac{e^{-2}(i - 2)}{20}$$

$$= \frac{i - 2}{20e^2}$$

$$(v) \text{ PV of } \int_{-\infty}^{\infty} \frac{e^{ix}}{(x^2 + 4)(x - 1)} dx = \pi i \operatorname{Res} [f(z); z = 1] + 2\pi i \operatorname{Res} [f(z); z = 2i]$$

$$\text{PV of } \int_{-\infty}^{\infty} \frac{\cos x + i \sin x}{(x^2 + 4)(x - 1)} dx = \pi i \left(\frac{e^i}{5} \right) + 2\pi i \left(\frac{i - 2}{20e^2} \right)$$

$$= \frac{\pi i}{5} (\cos 1 + i \sin 1) + \frac{\pi}{10e^2} (-1 - 2i)$$

$$= \frac{\pi}{5} (i \cos 1 - \sin 1) - \frac{\pi}{10e^2} (1 + 2i)$$

$$= -\frac{\pi}{5} \left(\sin 1 + \frac{1}{2e^2} \right) + i \left(\frac{\pi}{5} \cos 1 - \frac{2\pi}{10e^2} \right)$$

$$= -\frac{\pi}{5} \left(\sin 1 + \frac{1}{2e^2} \right) + i \frac{\pi}{5} \left(\cos 1 - \frac{1}{e^2} \right)$$

Comparing imaginary parts,

$$\text{PV of } \int_{-\infty}^{\infty} \frac{\sin x \, dx}{(x^2 + 4)(x - 1)} = \frac{\pi}{5} \left(\cos 1 - \frac{1}{e^2} \right)$$

Example 3

Using indentation along a branch cut, show that $\int_0^{\infty} \frac{\sin x}{x} dx = \frac{\pi}{2}$.

Solution

(i) Let $\int_C f(z) dz = \int_C \frac{e^{iz}}{z} dz$

(ii) Let $f(z) = \frac{e^{iz}}{z}$

The pole is given by
 $z = 0$

(iii) The pole $z = 0$ lies on the real axis (Fig. 5.30).

(iv) $\text{Res}[f(z); z = 0] = \lim_{z \rightarrow 0} z f(z)$
 $= \lim_{z \rightarrow 0} e^{iz}$
 $= e^0$
 $= 1$

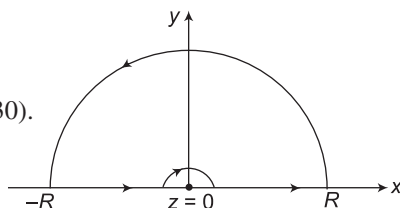


Fig. 5.30

(v) $\int_{-\infty}^{\infty} \frac{e^{ix}}{x} dx = \pi i \text{Res}[f(z); z = 0]$

$$\int_{-\infty}^{\infty} \frac{\cos x + i \sin x}{x} dx = \pi i (1)$$

Comparing imaginary parts,

$$\int_{-\infty}^{\infty} \frac{\sin x}{x} dx = \pi$$

$$2 \int_0^{\infty} \frac{\sin x}{x} dx = \pi \quad \left[\because \int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx, \text{ if } f(-x) = f(x) \right]$$

$$\int_0^{\infty} \frac{\sin x}{x} dx = \frac{\pi}{2}$$

EXERCISE 5.4

Evaluate the following integrals using contour integration.

$$1. \int_0^{\infty} \frac{1 - \cos x}{x^2} dx$$

$$\left[\text{Ans.: } \frac{\pi}{2} \right]$$

$$2. \int_{-\infty}^{\infty} \frac{x}{8 - x^3} dx$$

$$\left[\text{Ans.: } -\frac{\pi\sqrt{3}}{6} \right]$$

$$3. \int_0^{\infty} \frac{\sin mx}{x} dx, m > 0$$

$$\left[\text{Ans.: } \frac{\pi}{2} \right]$$

$$4. \int_{-\infty}^{\infty} \frac{\cos mx}{x} dx$$

$$\left[\text{Ans.: } 0 \right]$$

$$5. \int_{-\infty}^{\infty} \frac{dx}{x^2 - ix}$$

$$\left[\text{Ans.: } \pi \right]$$

$$6. \int_{-\infty}^{\infty} \frac{x^2}{x^4 - 1} dx$$

$$\left[\text{Ans.: } \frac{\pi}{2} \right]$$

$$7. \int_{-\infty}^{\infty} \frac{dx}{x^4 + 3x^2 - 4}$$

$$\left[\text{Ans.: } -\frac{\pi}{10} \right]$$

$$8. \int_{-\infty}^{\infty} \frac{3x+2}{x(x-4)(x^2+9)} dx$$

$$\left[\text{Ans.: } -\frac{14}{75}\pi \right]$$

Points to Remember

Evaluation of a Real Definite Integral of a Rational Function of $\cos \theta$ and $\sin \theta$

$$\int_0^{2\pi} f(\sin \theta, \cos \theta) d\theta = 2\pi i \text{ (sum of residues at all poles lying inside the circle } |z| = 1)$$

Evaluation of Improper Real Integral of a Rational Function

$$\int_{-\infty}^{\infty} f(x) dx = 2\pi i \text{ (sum of residues at poles in the upper half of the } z\text{-plane)}$$

where $f(x) = \frac{P(x)}{Q(x)}$ and the degree of $Q(x)$ is greater than the degree of $P(x)$ by at least 2 and $Q(x)$ has no real roots.

Evaluation of Improper Real Integral of a Rational Function Including Trigonometric Functions

$$\int_{-\infty}^{\infty} \frac{\cos mx}{Q(x)} dx = 2\pi i \text{ (sum of residues at poles in the upper half of the } z\text{-plane)}$$

$$\int_{-\infty}^{\infty} \frac{\sin mx}{Q(x)} dx = 2\pi i \text{ (sum of residues at poles in the upper half of the } z\text{-plane)}$$

where $Q(x)$ is a polynomial in x .

Evaluation of Improper Integral When Simple Poles Lie on the Real Axis

If $f(z)$ has a simple pole $z = b$ on the real axis and another simple pole $z = z_1$ in the upper half of the z -plane then

$$\int_{-\infty}^{\infty} f(x) dx = \pi i \operatorname{Res} [f(z); z = b] + 2\pi i \operatorname{Res} [f(z); z = z_1]$$